ENCLUSTRA
FPGA SOLUTIONS

Enclustra User Schematic

Andromeda XZU90 Revision 2.1

Disclaimer

User schematic do not include proprietary Enclustra design elements, such as power supply circuits.
These circuits use optimized designs which are extremely compact and remain proprietary to Enclustra.
The user manual provides all necessary information to use the module and its interfaces.

User schematic are provided after purchase of Enclustra hardware, and may also be provided in certain cases before purchase, to assist in product evaluation.
No part of the schematic may be copied or modified without written permission from Enclustra.

Full schematic, including a full bill-of-materials, may be available through the purchase of a hardware licence for the product in question.
Please contact Enclustra Sales for more information.

Note: DNE = Do Not Equip (parts not equipped by default)

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Andromeda XZU90 FPGA SoC Module

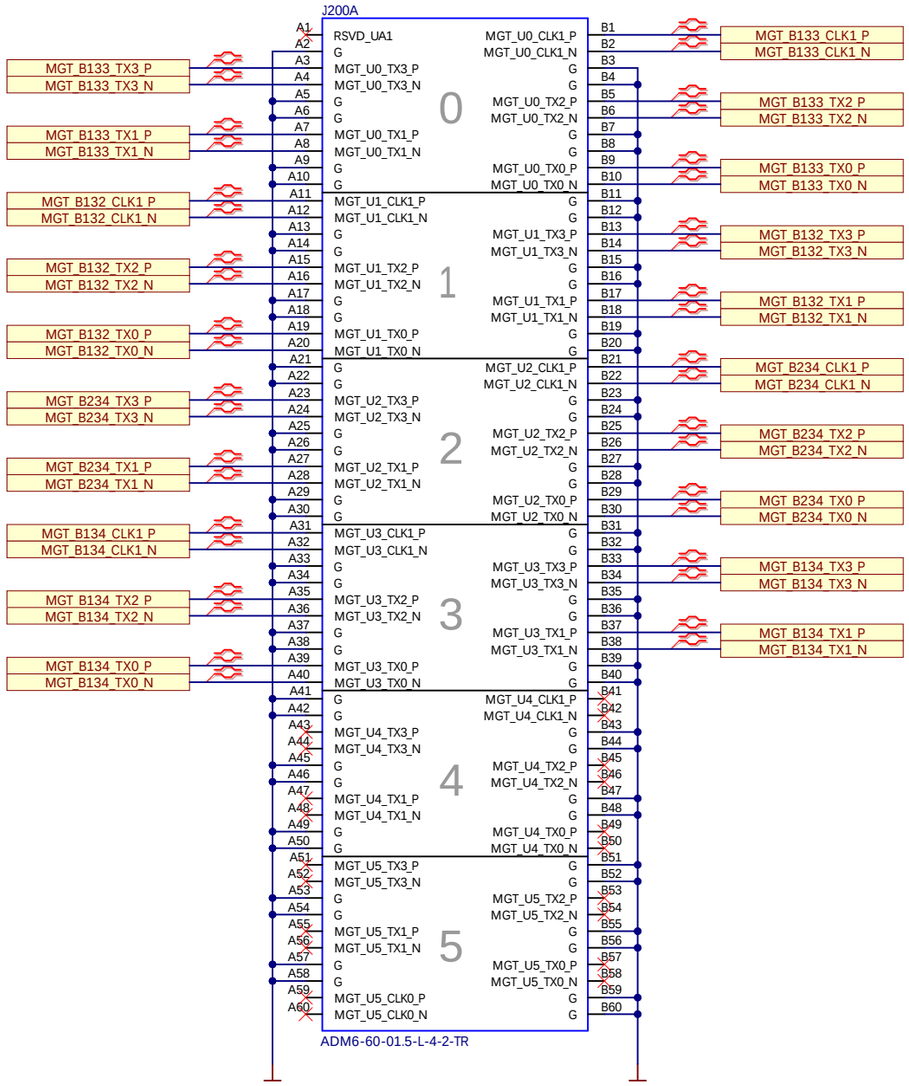
Revision 2.1

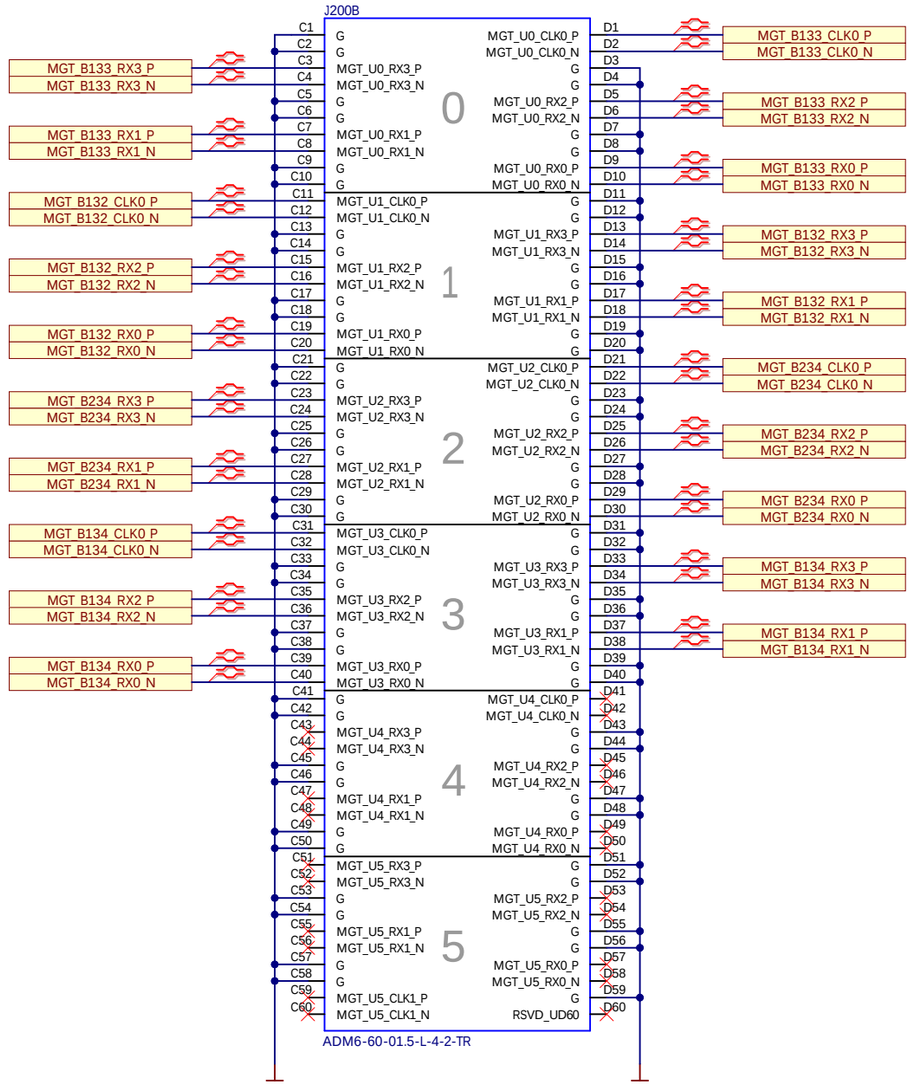
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Sheet 04: Module Connector W
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Sheet 24: PL Converters INT | INT_IO | 1.8V | 3.3V
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Connector U

GTY MGT - GTY MGT - GTH MGT - GTY MGT - NC - NC

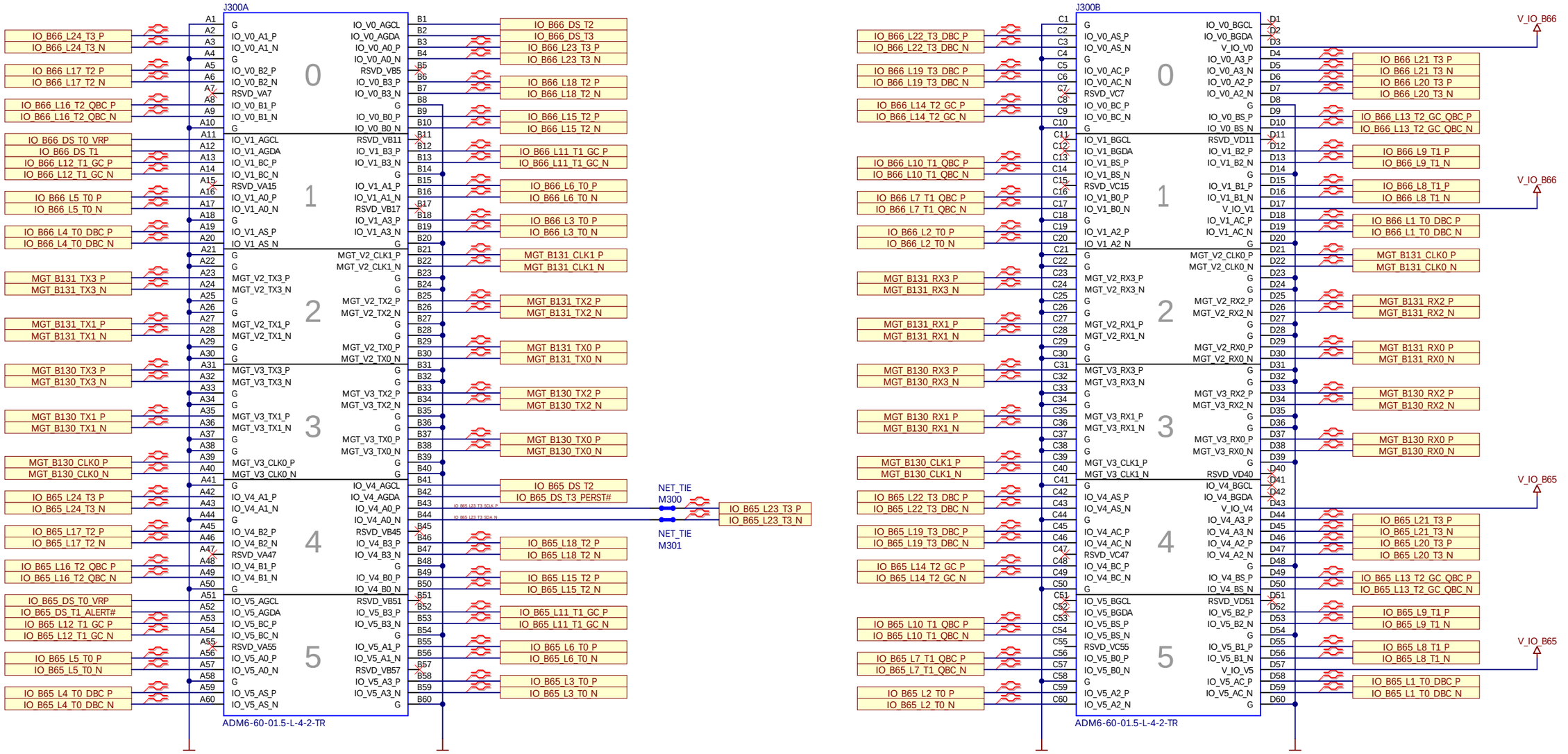




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Connector V

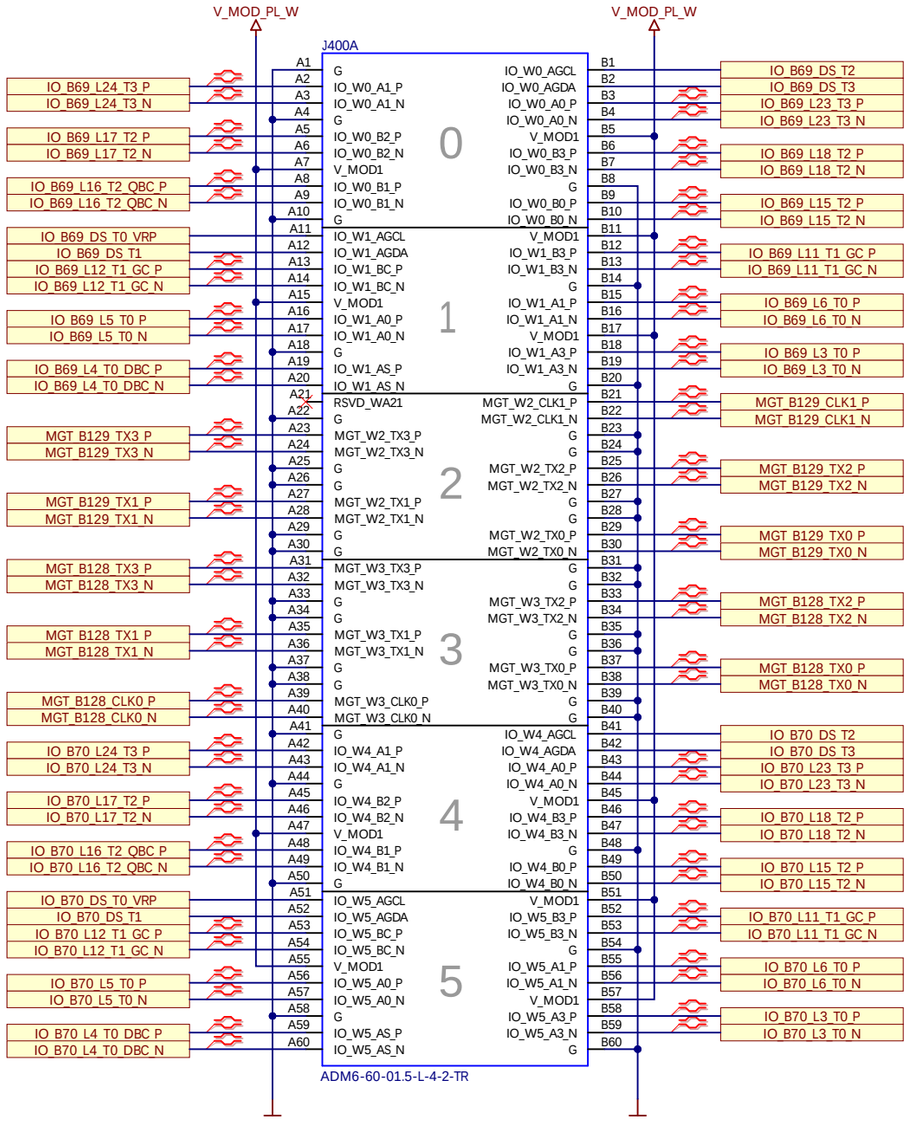
HP IO - HP IO - GTY MGT - GTY MGT - HP IO - HP IO

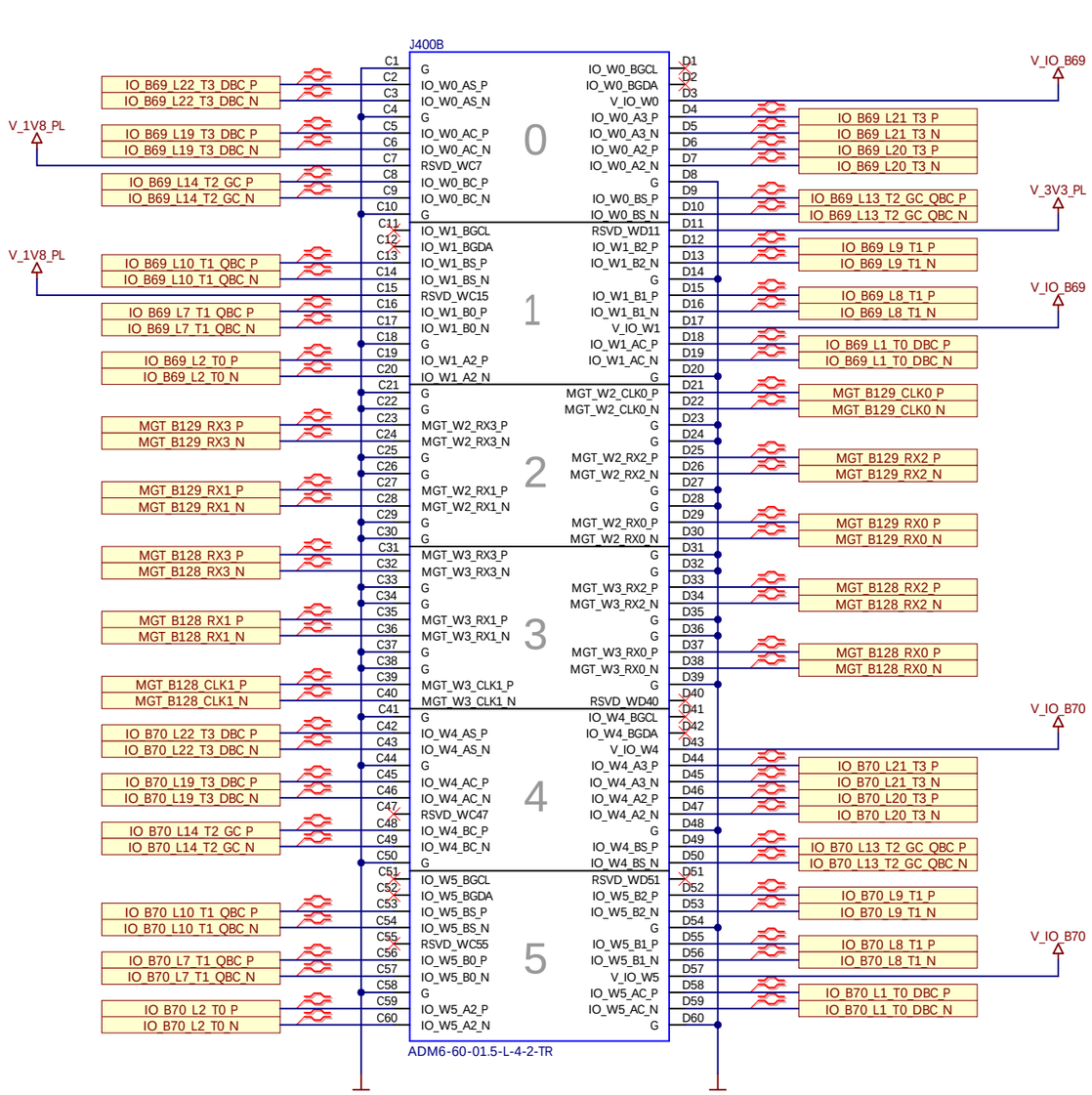


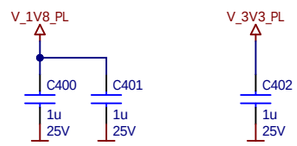
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Connector W

HP IO - HP IO - GTY MGT - GTY MGT - HP IO - HP IO



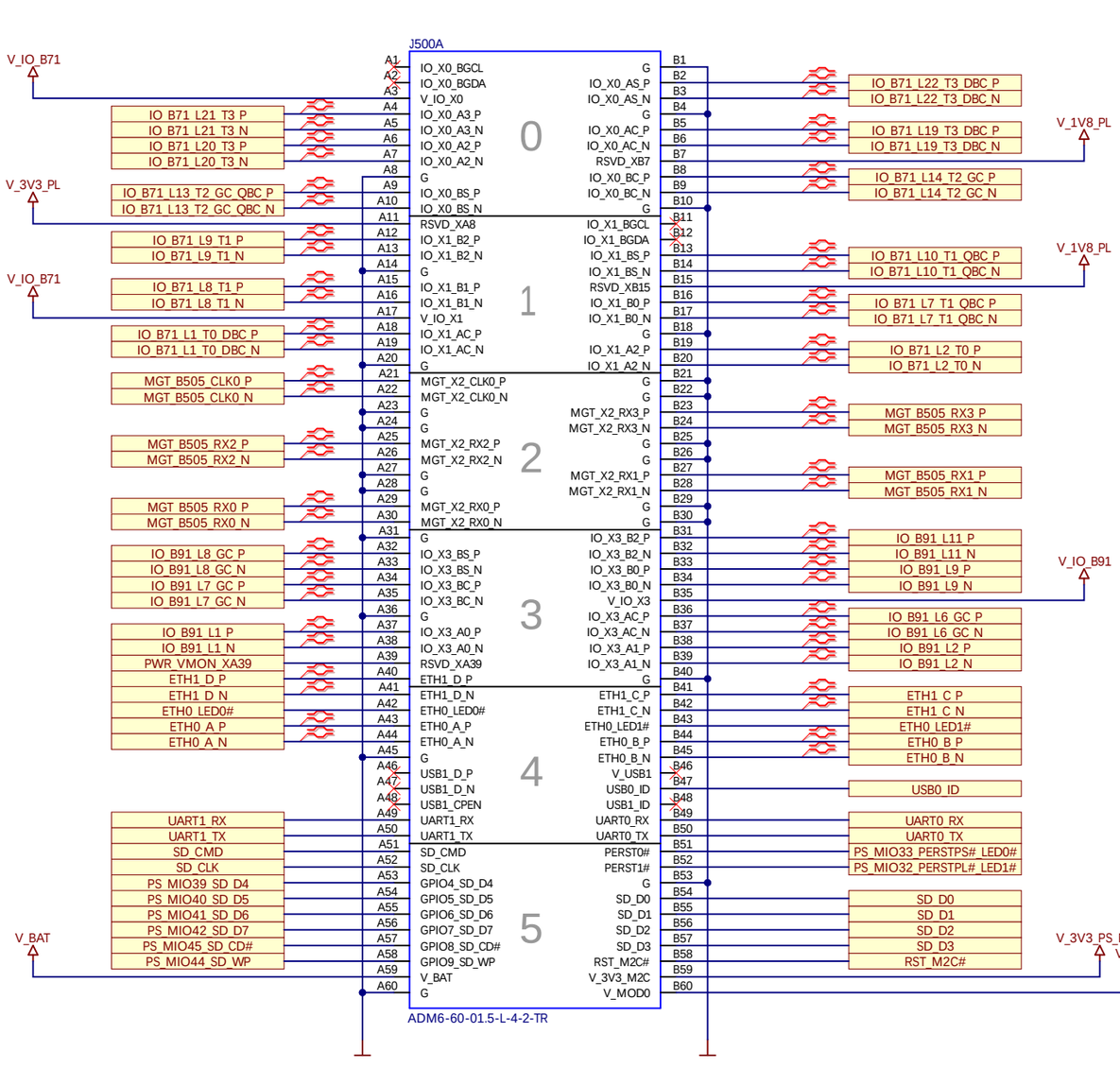


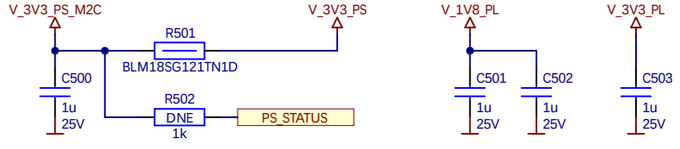


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Connector X

HP IO - HP IO - GTR MGT - HD IO - ETH/USB - Control

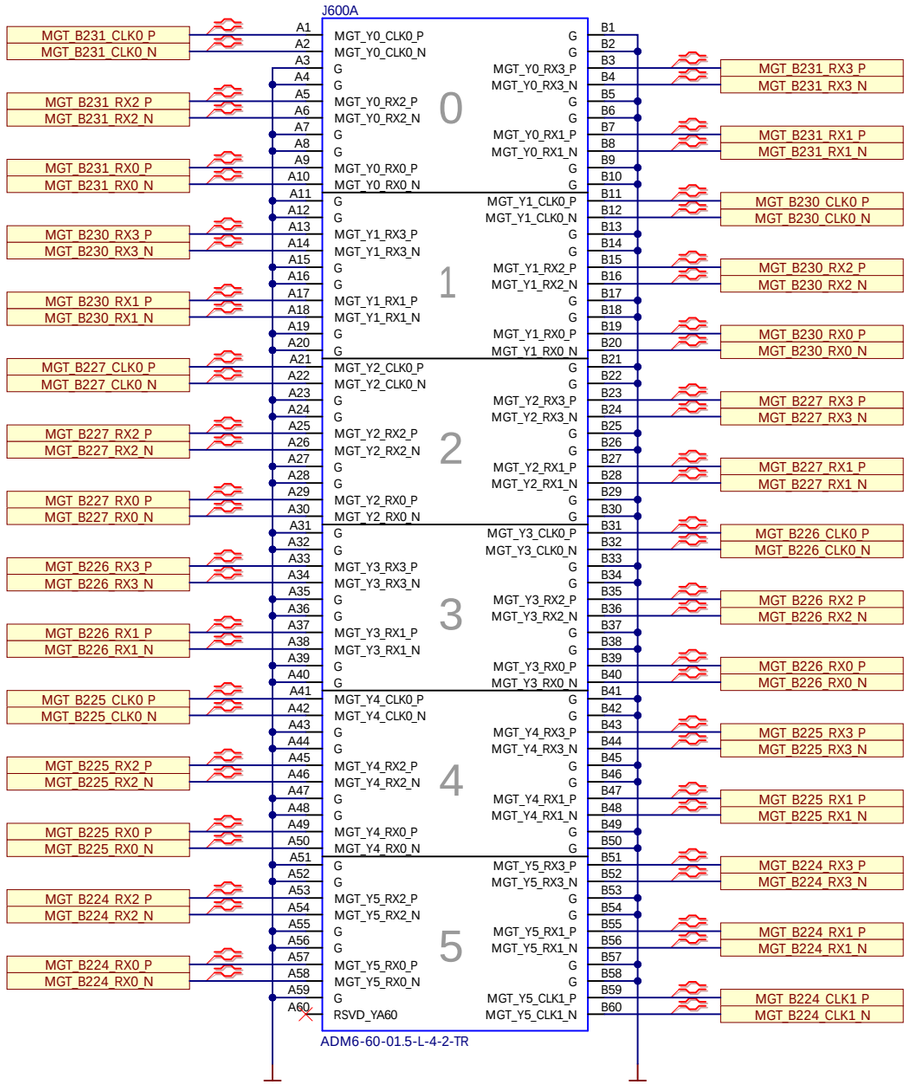


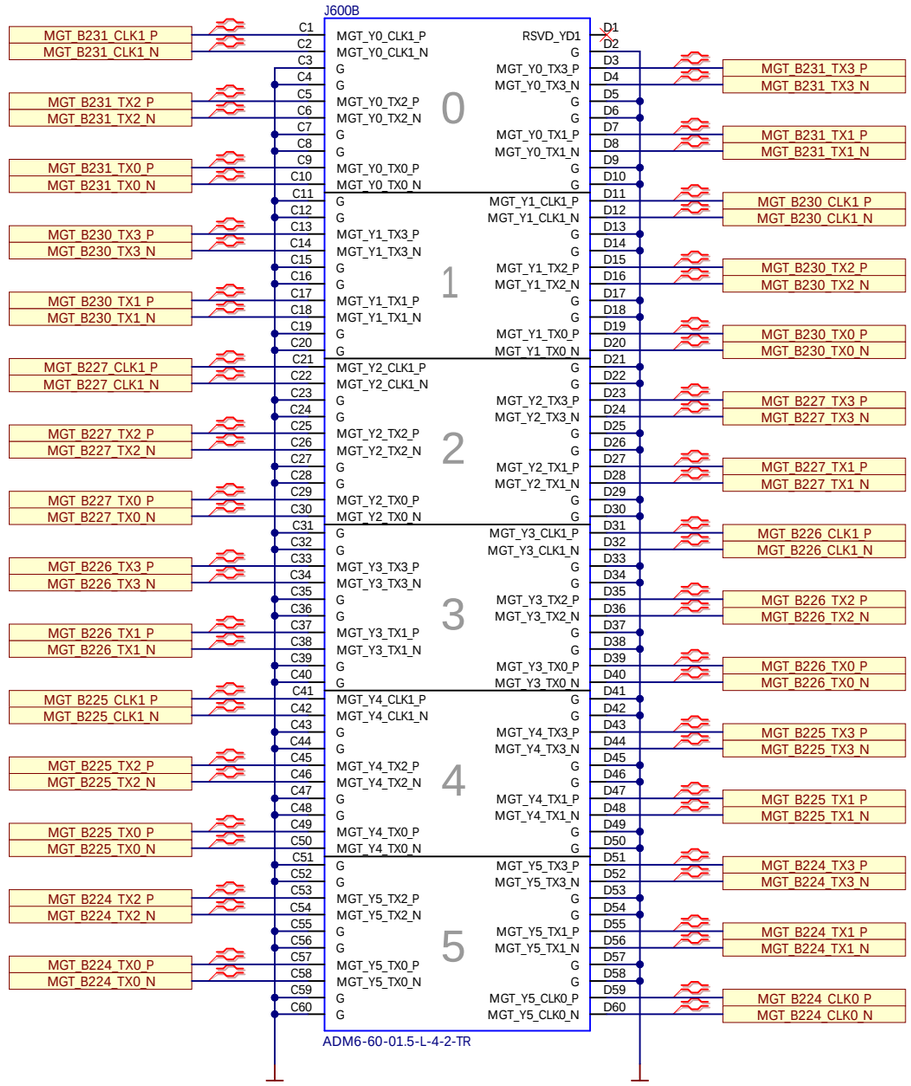


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Connector Y

GTH MGT - GTH MGT - GTH MGT - GTH MGT - GTH MGT - GTH MGT

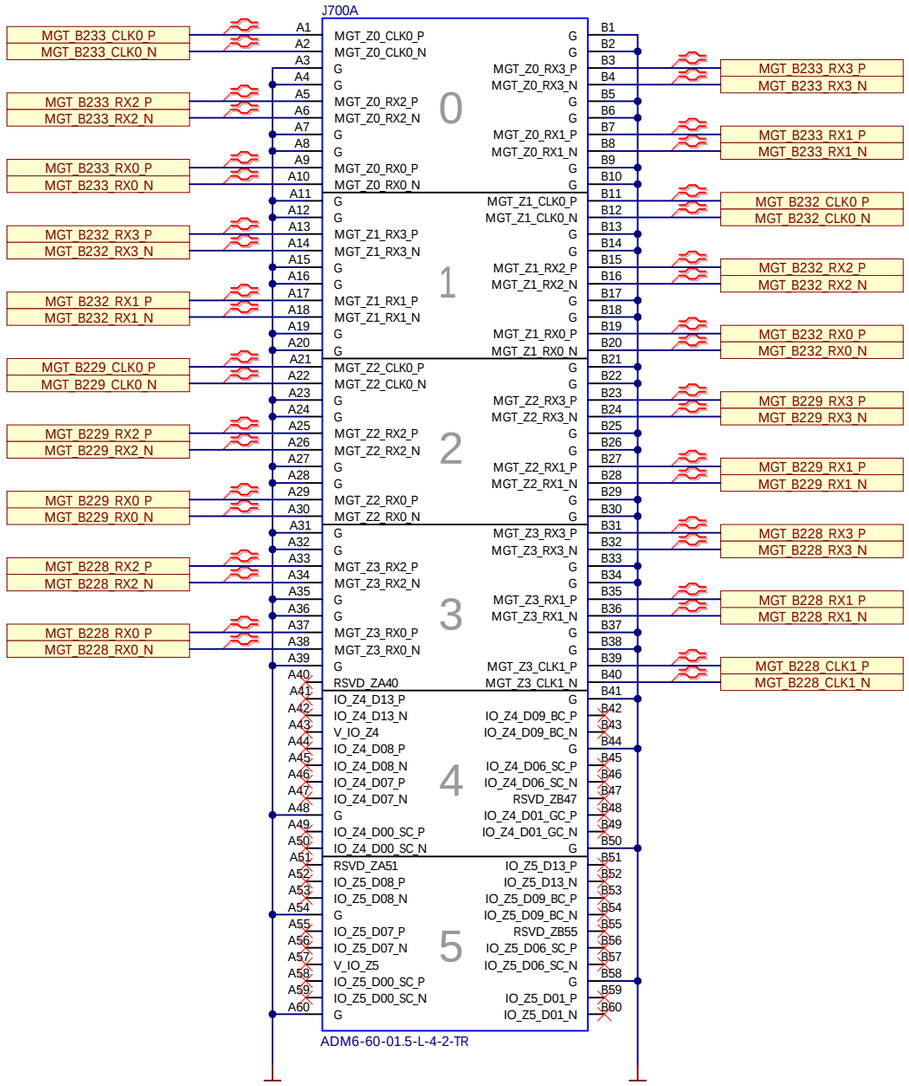


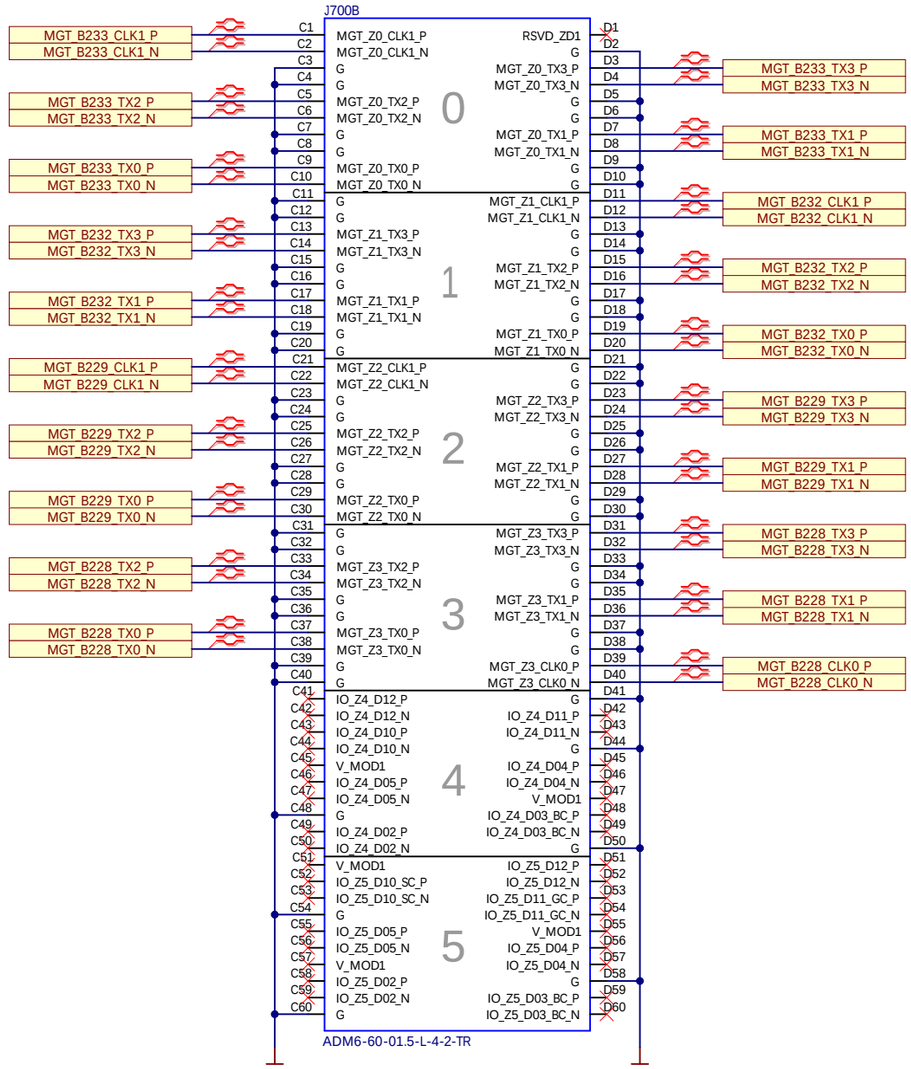


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Connector Z

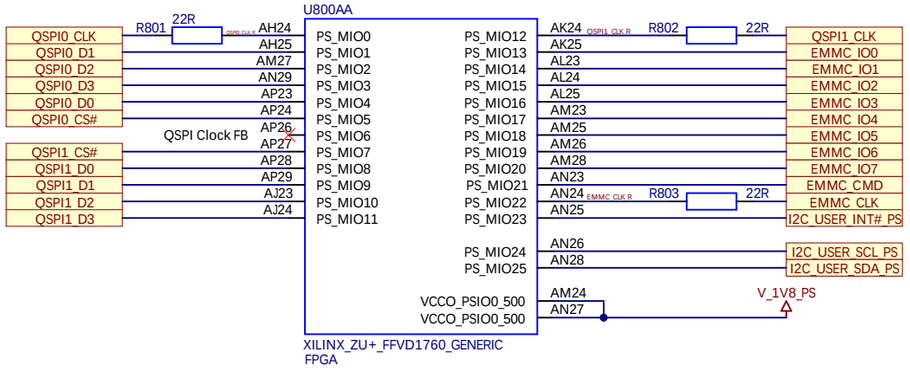
GTH MGT - GTH MGT - GTH MGT - GTH MGT - NC - NC

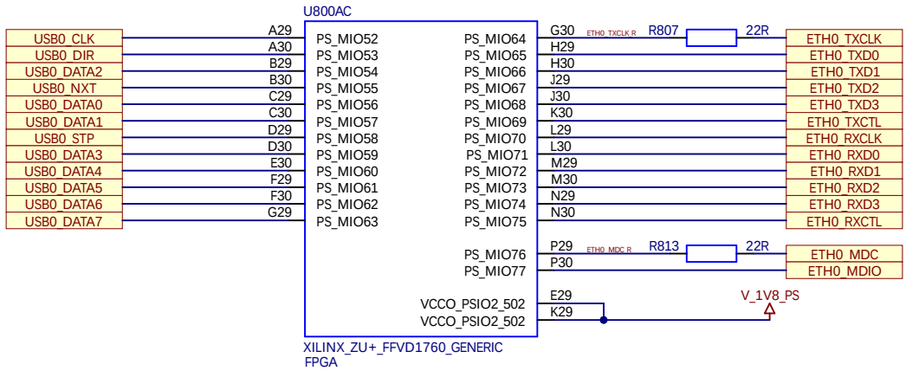




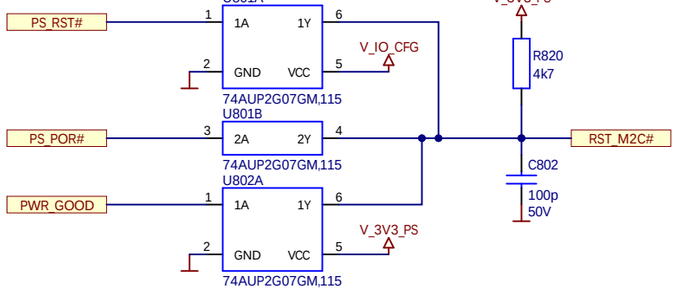
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PS MIO Bank 500 & 502

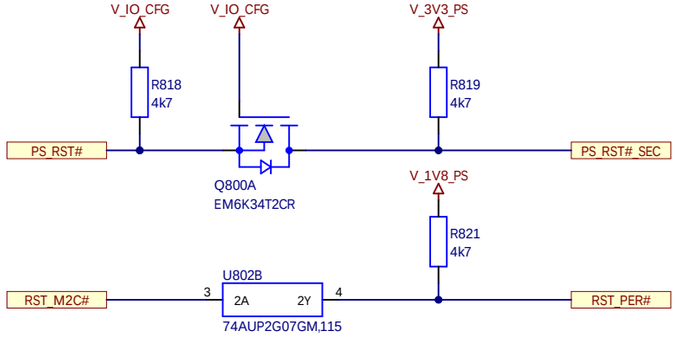




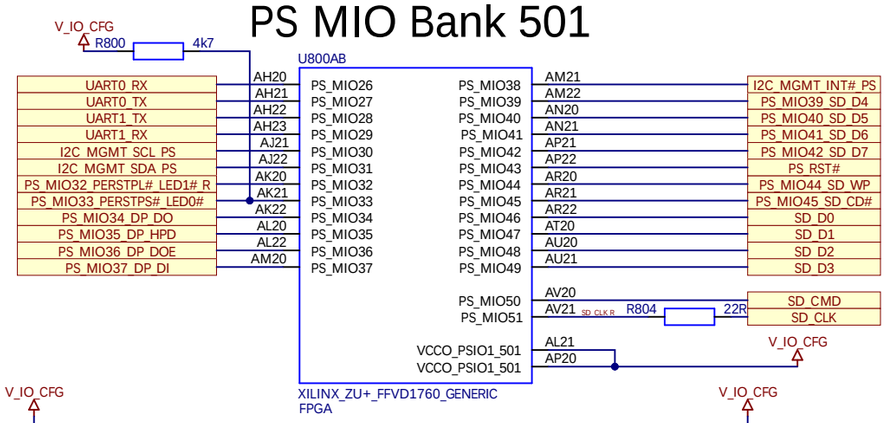
PS Reset Generation



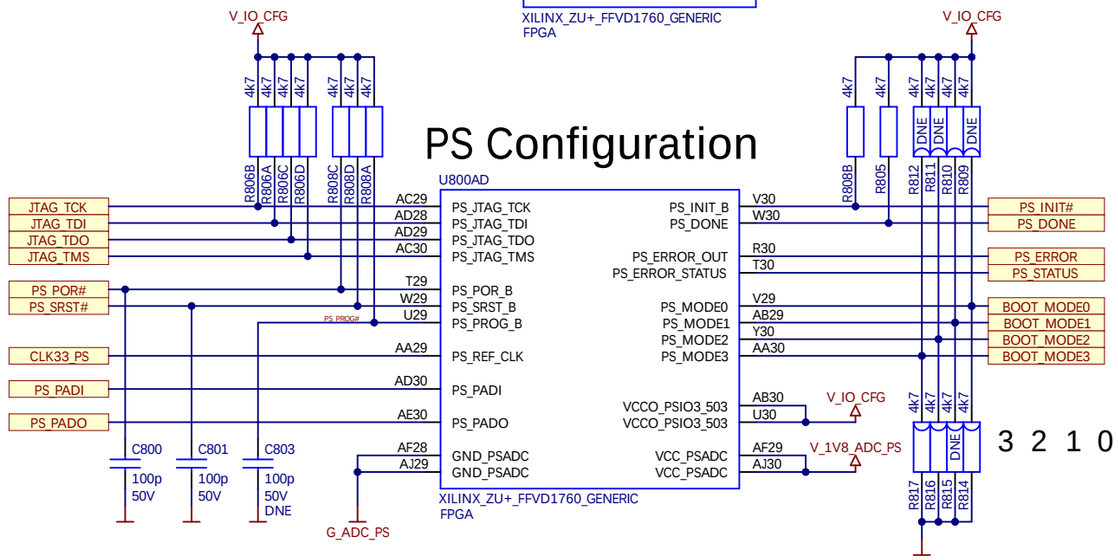
PS Peripheral Reset



PS MIO Bank 501

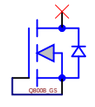


PS Configuration

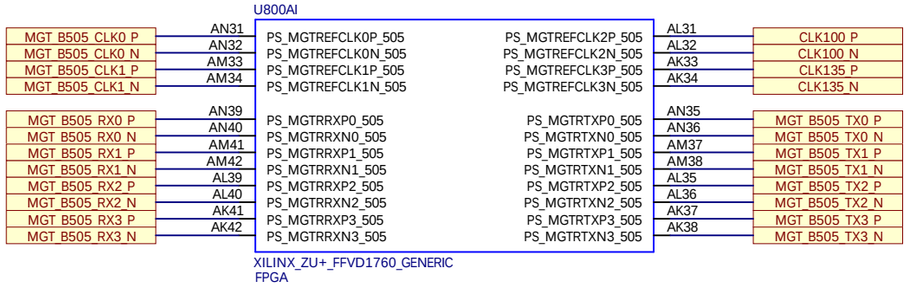


Supported Boot Modes

Boot Mode	Mode Pins [3:0]
PS JTAG	0000
QSPI (default)	0010
SD 2.0	0101
EMMC	0110
SD 3.0	1110

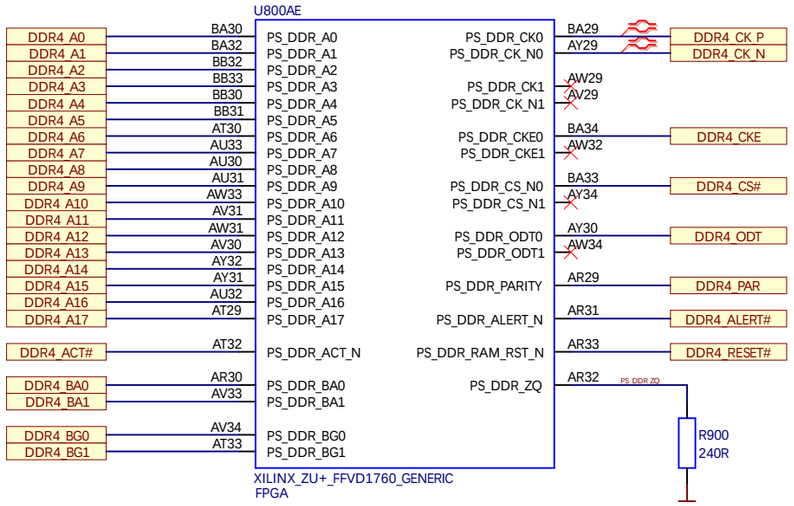


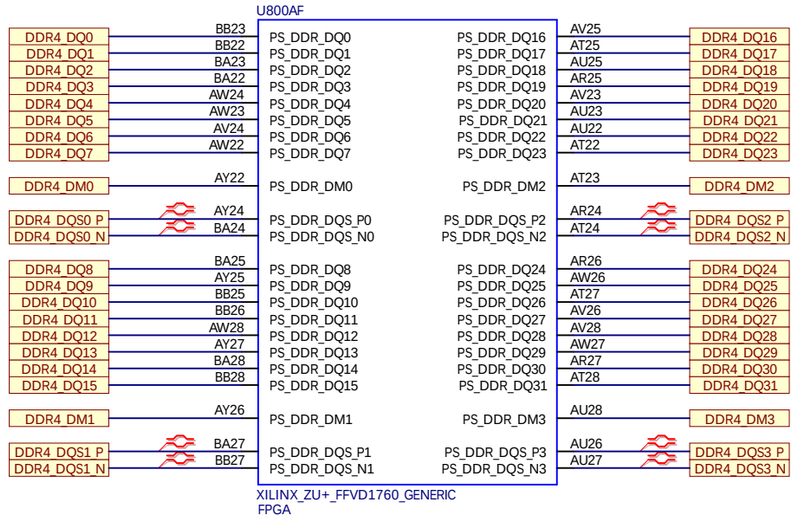
PS GTR Transceivers



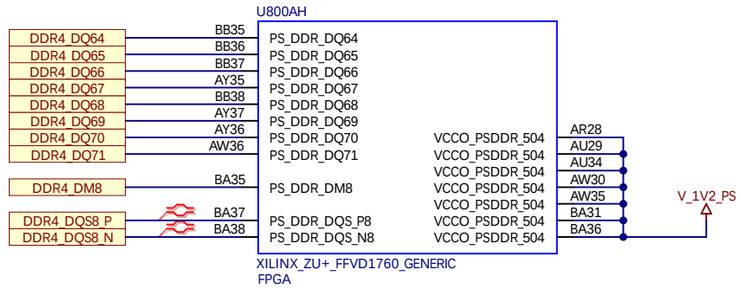
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PS DDR4 Interface

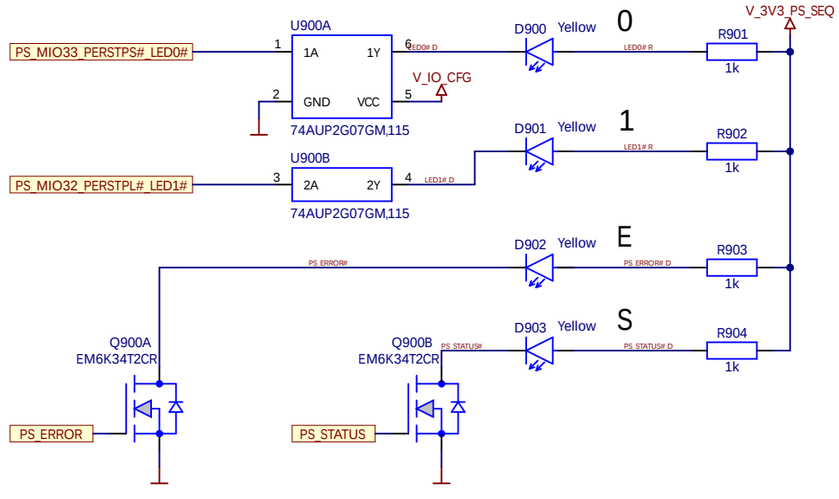






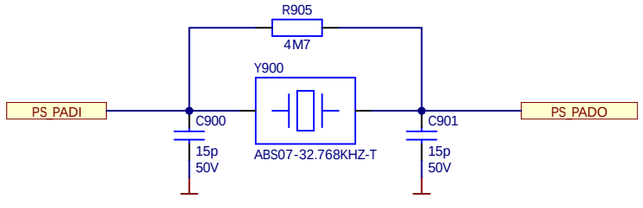


PS LEDs



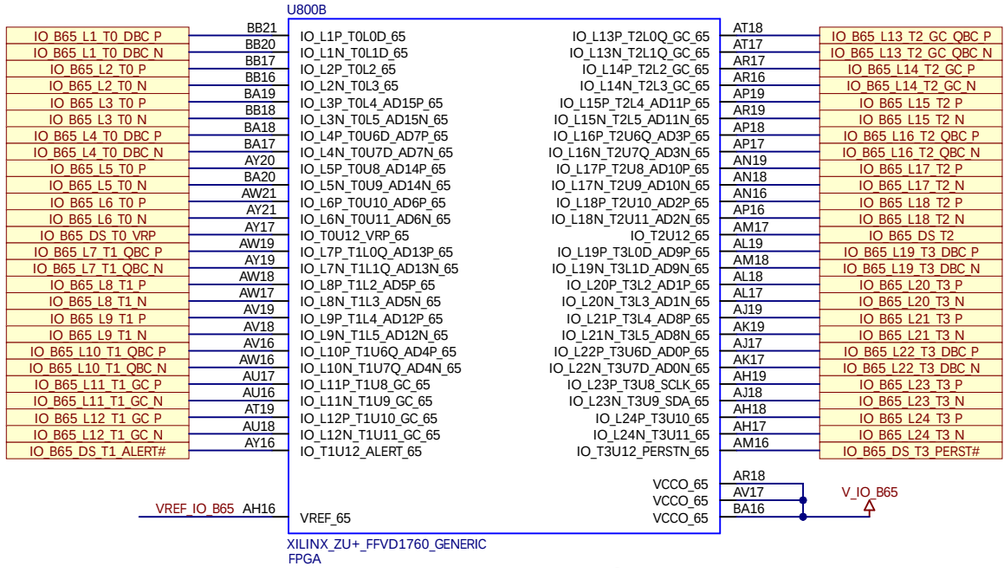
PS RTC Crystal

32.768 kHz

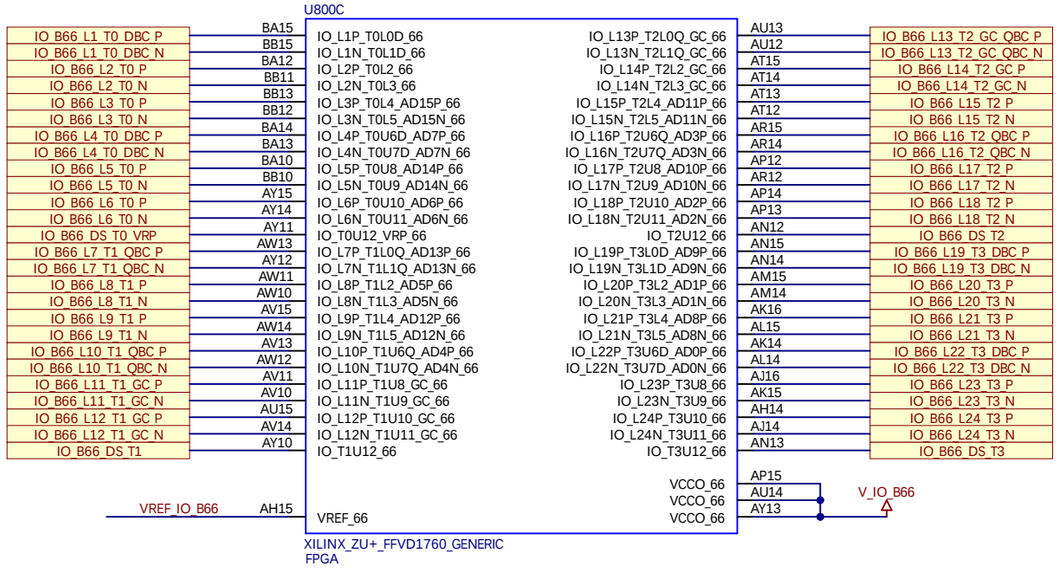


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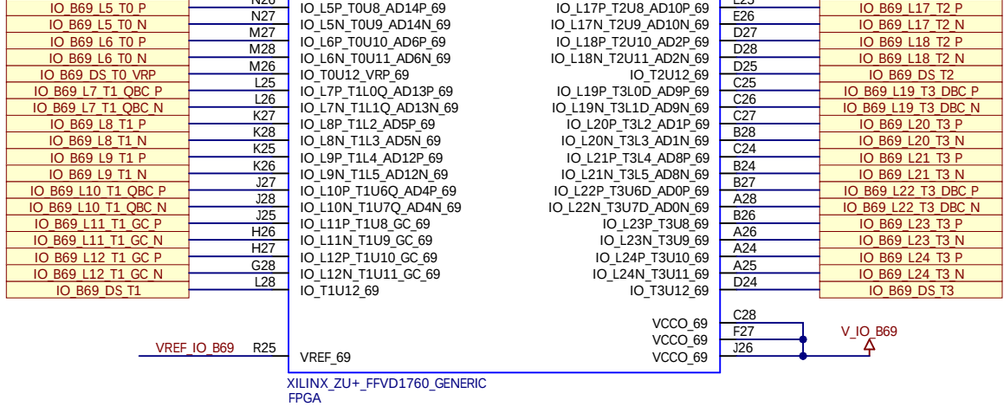
FPGA HP Bank 65



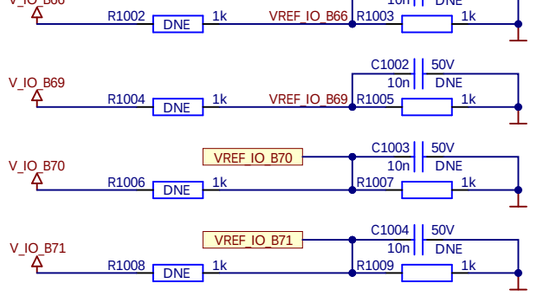
FPGA HP Bank 66



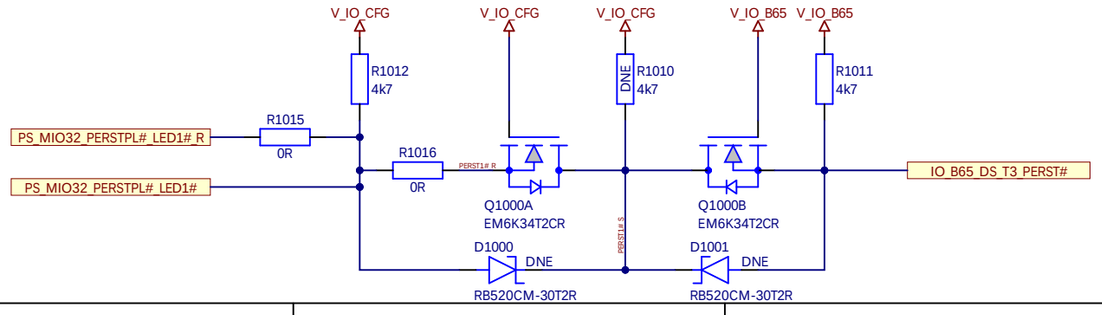
FPGA HP Bank 69



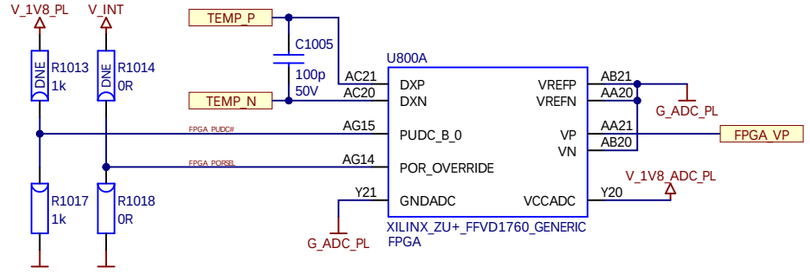
PL VREF Generation



PL PERST# Level Shifter

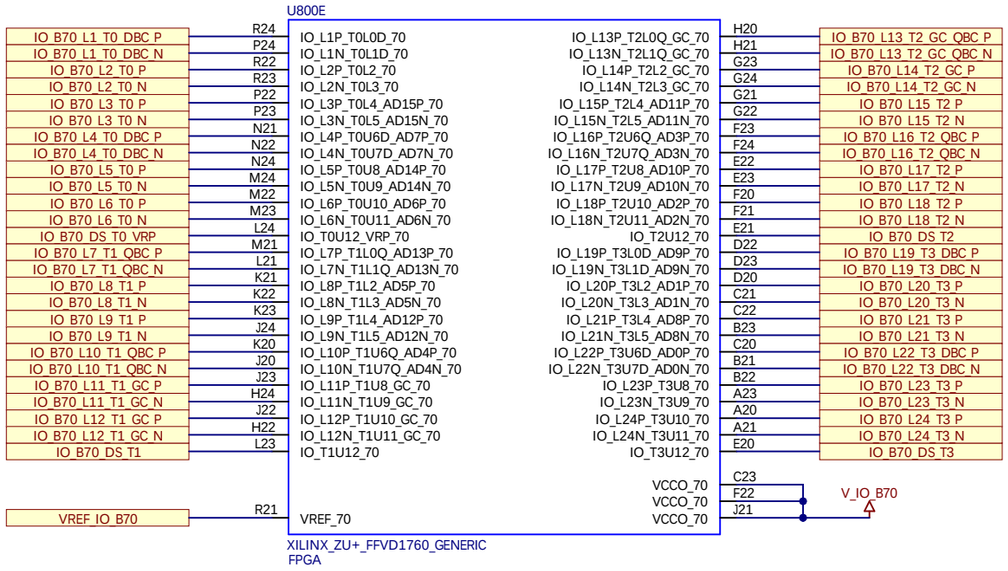


PL System Monitor

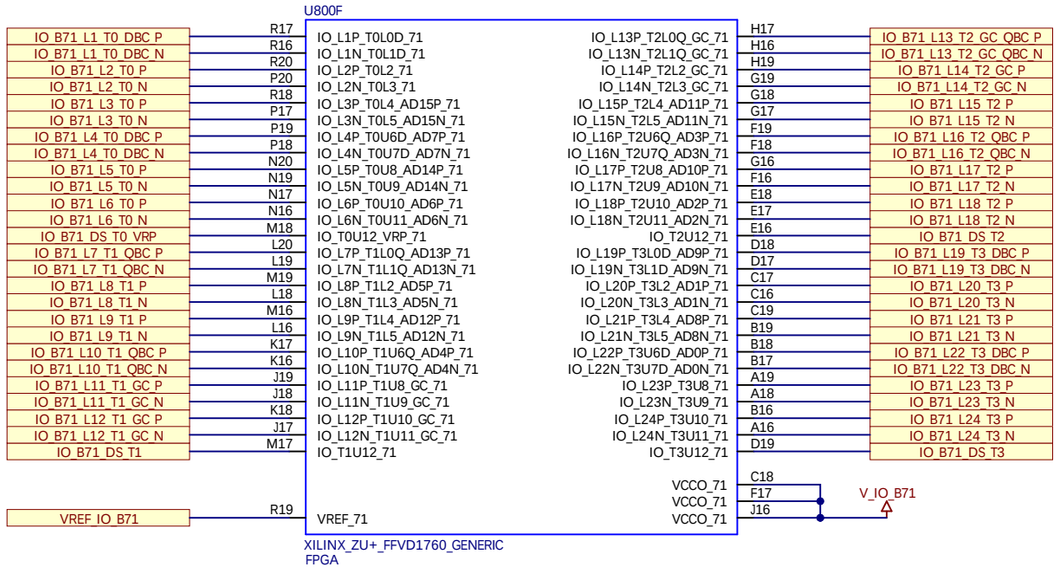


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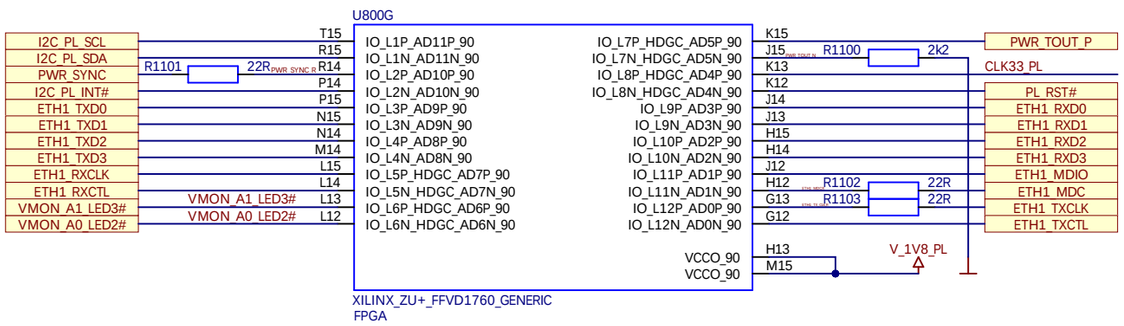
FPGA HP Bank 70



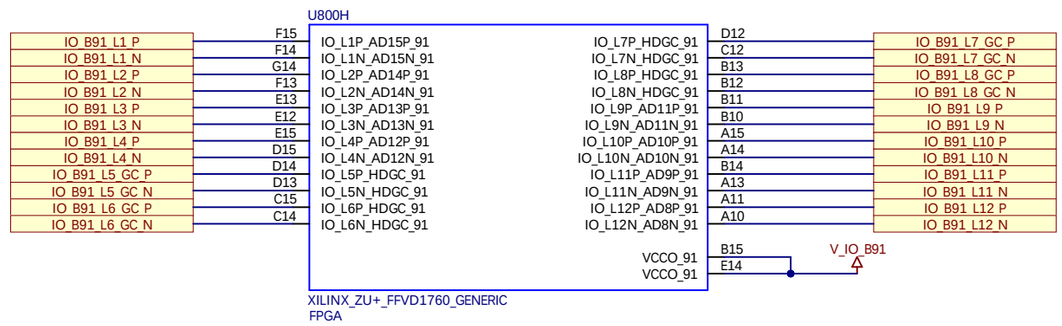
FPGA HP Bank 71



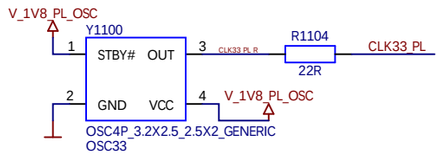
FPGA HD Bank 90



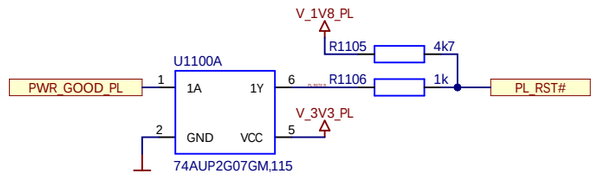
FPGA HD Bank 91



PL Oscillator 33.3 MHz



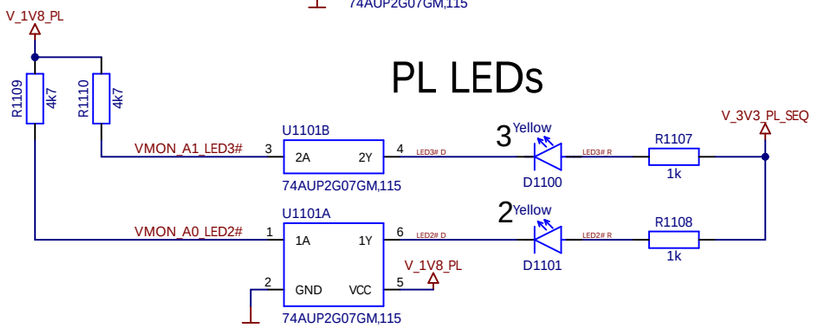
PL Reset Generation



Oscillator Power Filter

not included in user schematics

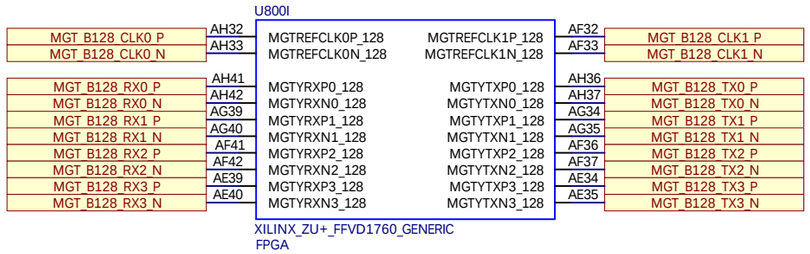
PL LEDs

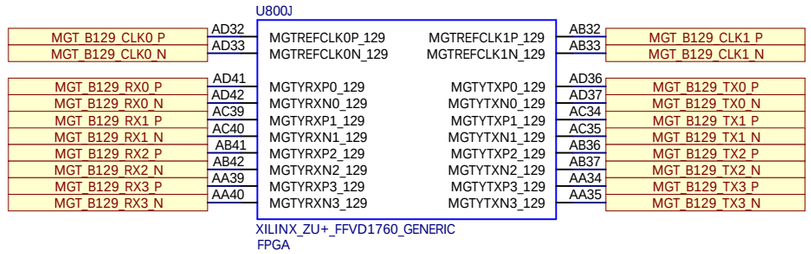


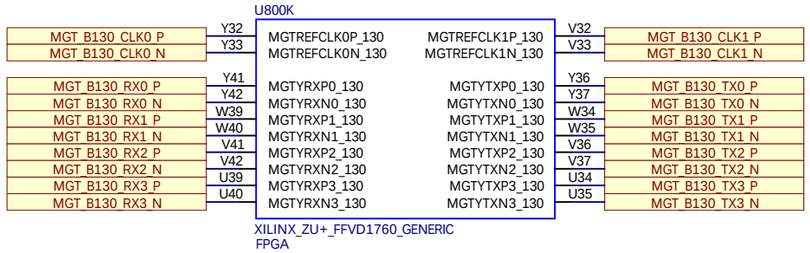
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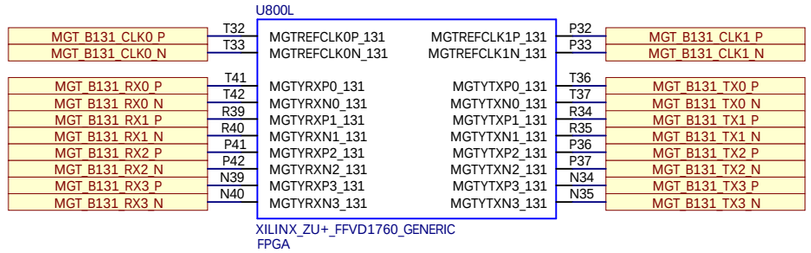
FPGA MGTY

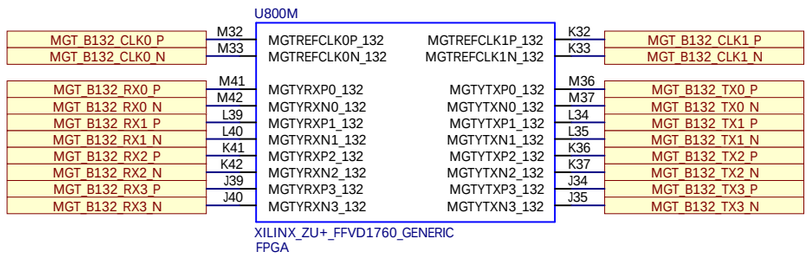
B128 - B129 - B130 - B131 - B132 - B133 - B134

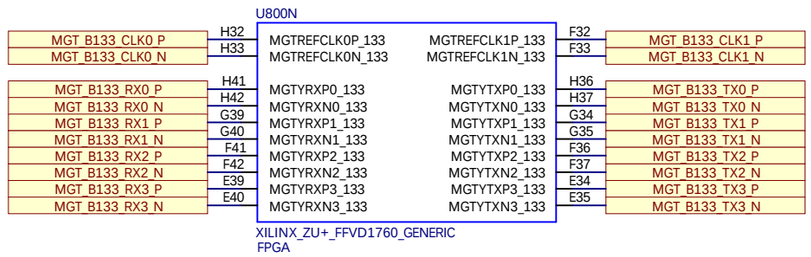


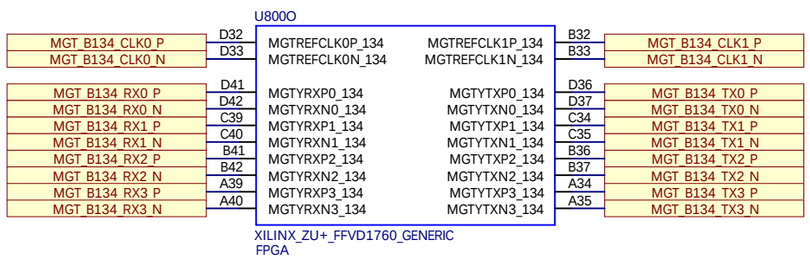








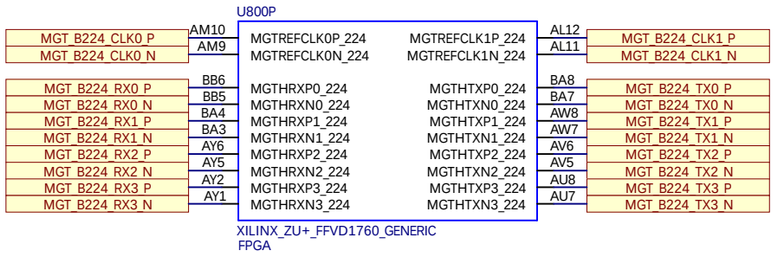


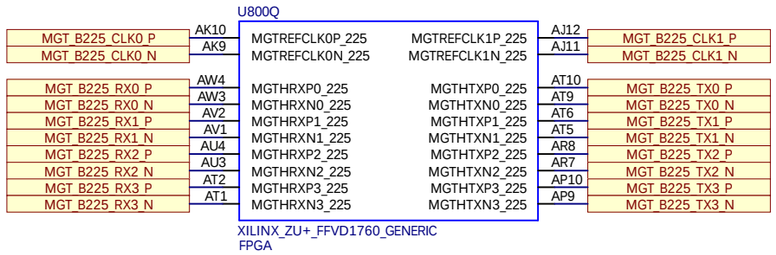


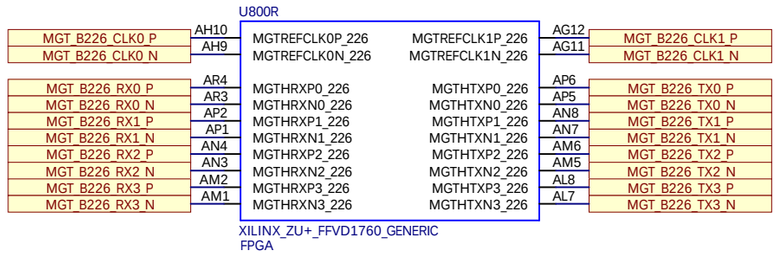
ENCLUSTRA FPGA SOLUTIONS	Project AM-XZU90.PrjPcb	Sheet Name AM-XZU90_P12_FPGA_MGTY.SchDoc	Customer No 0000	Revision R2.1	DNE = Do Not Equip	Confidential
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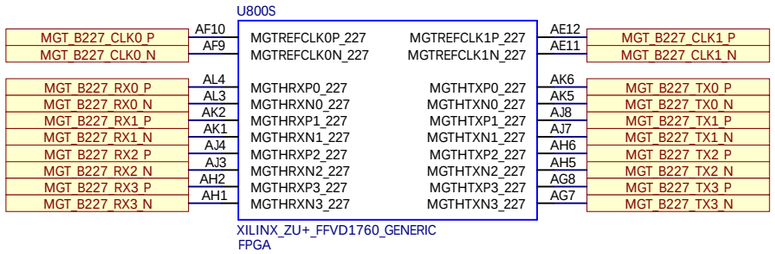
FPGA MGTH

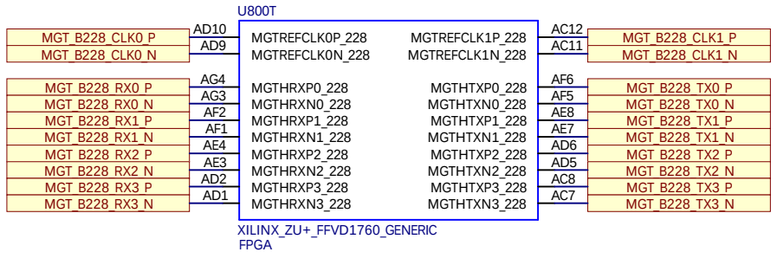
B224 - B225 - B226 - B227 - B228 - B229 - B230 - B231 - B232 - B233 - B234

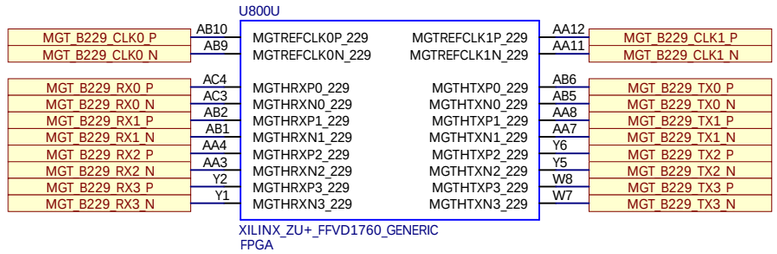


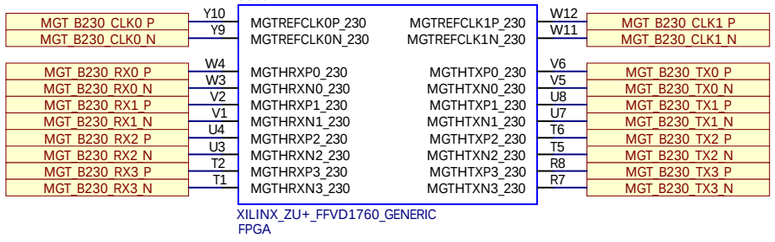


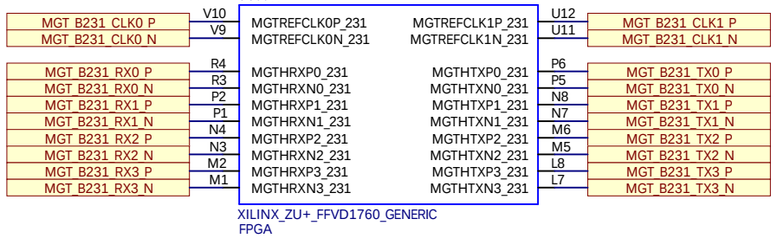


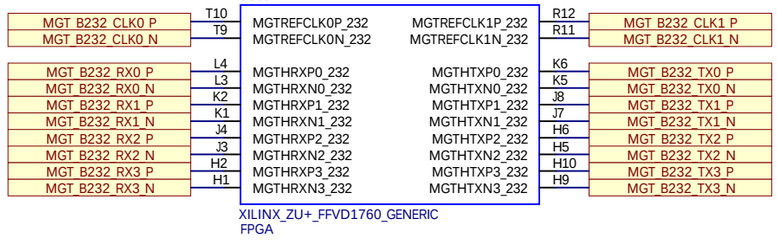


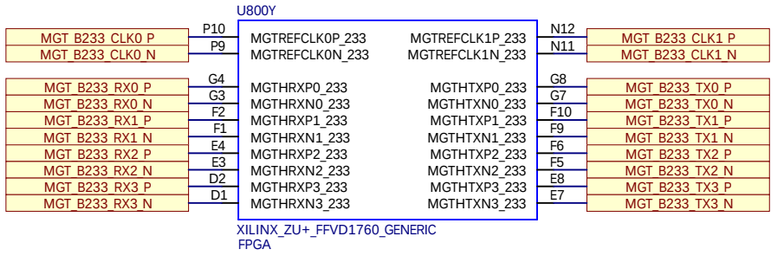


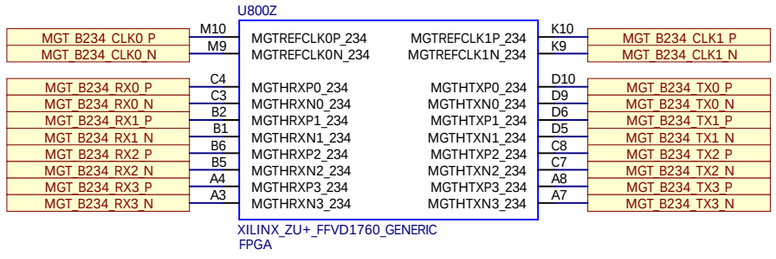






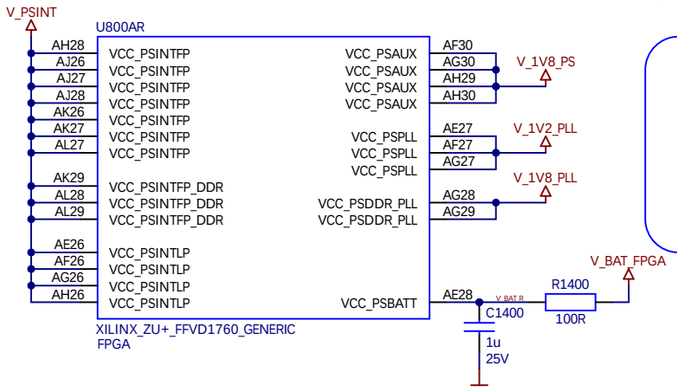






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FPGA PS Power PS System Monitor Power



not included in user schematics

PS PLL Power Filter

not included in user schematics

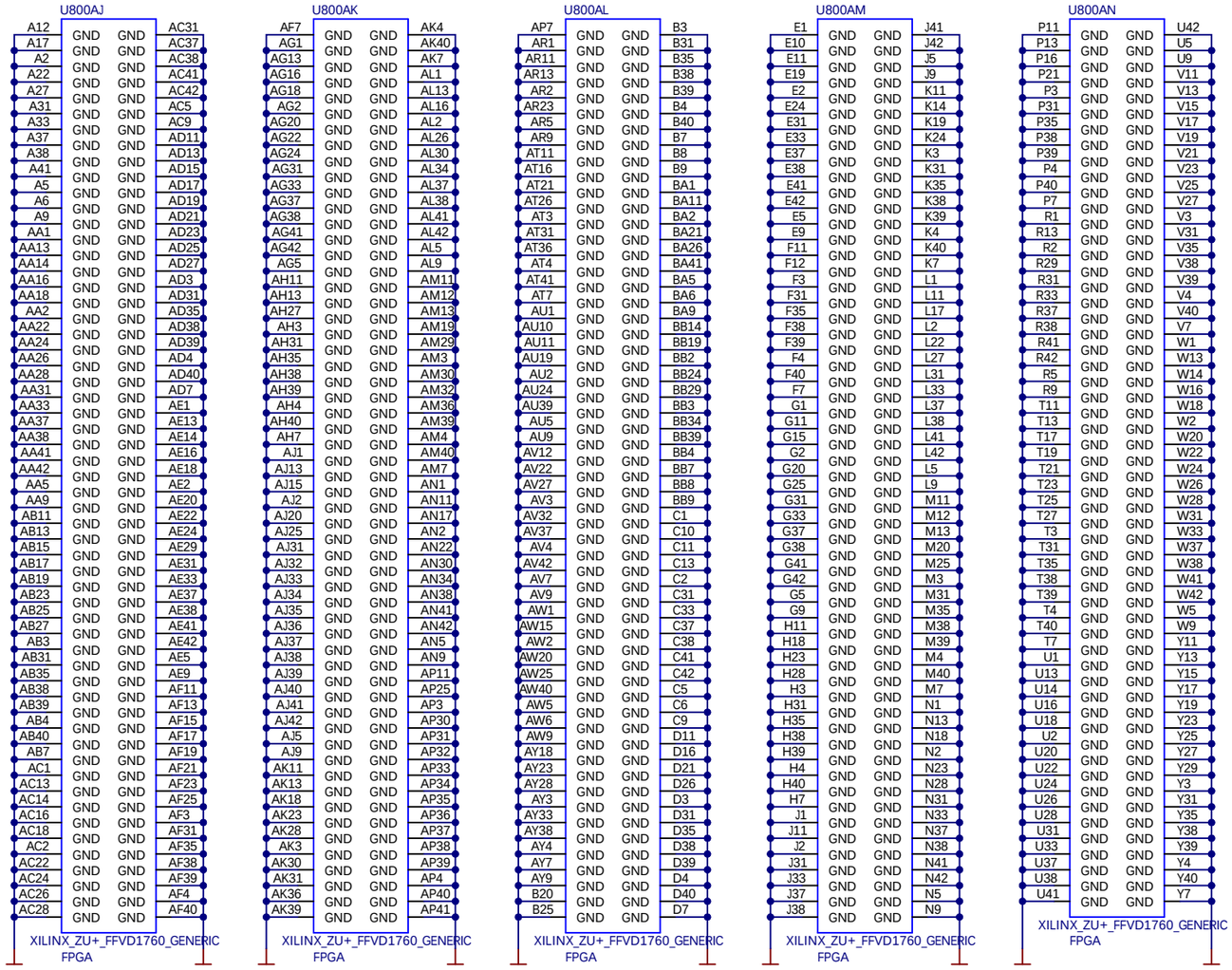
FPGA PS Decoupling

not included in user schematics

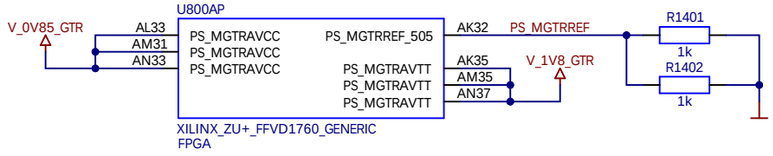
FPGA PS IO Decoupling

not included in user schematics

FPGA Ground



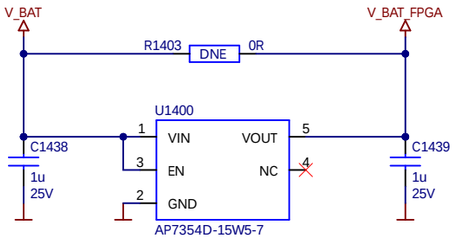
FPGA PS MGT Power



PS MGT Power Filter

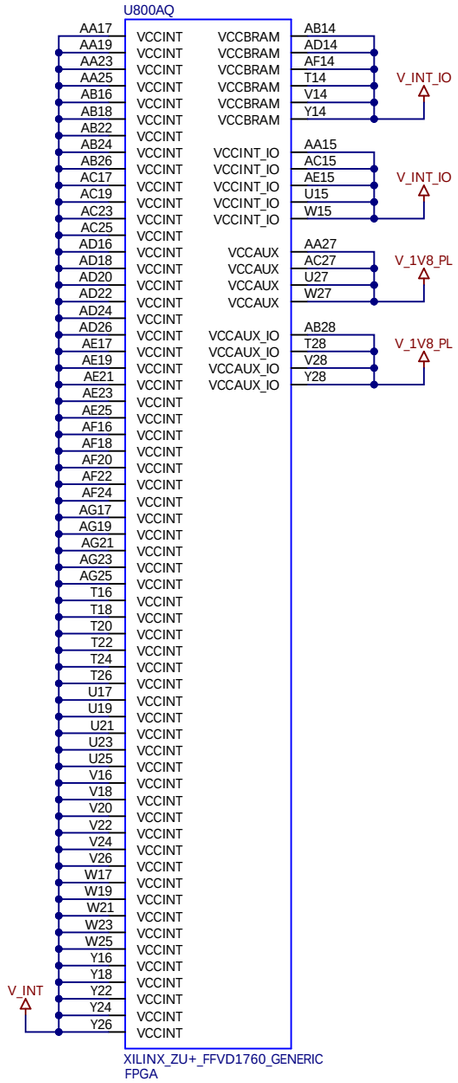
not included in user schematics

LDO 1.5V 150mA

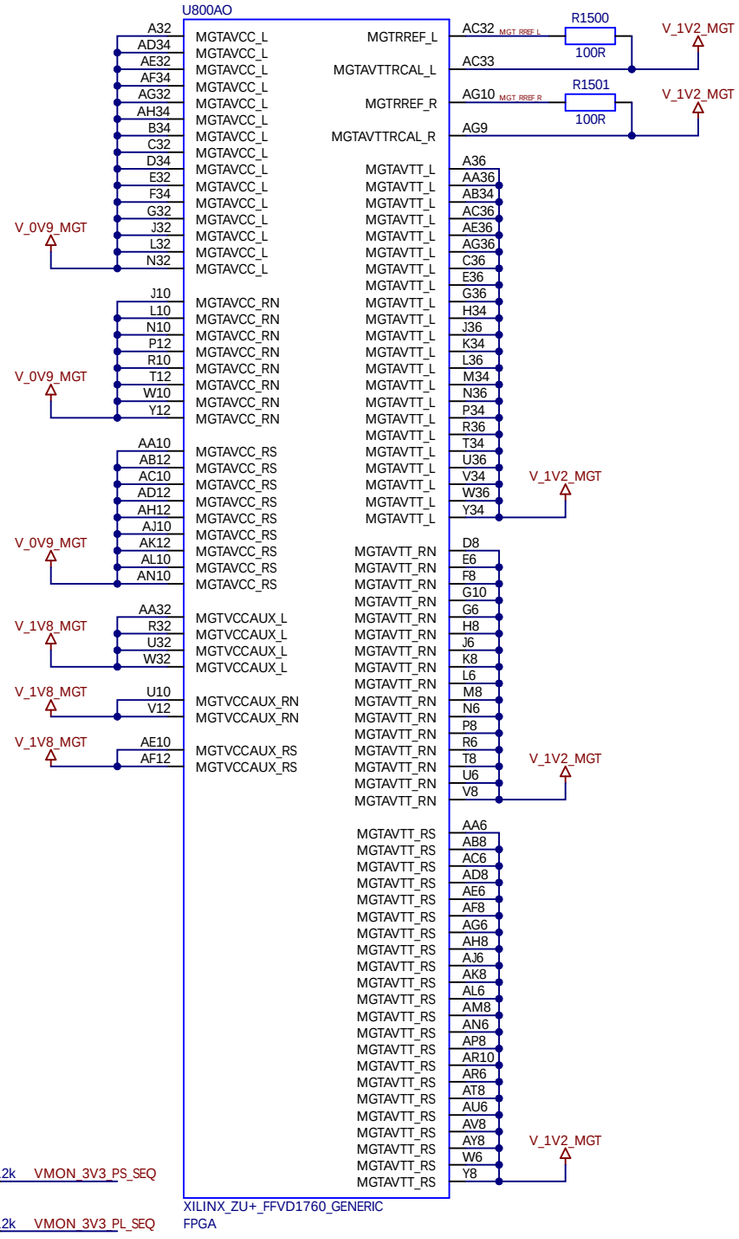


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FPGA Power



FPGA MGT Supplies



FPGA PL MGT Decoupling

not included in user schematics

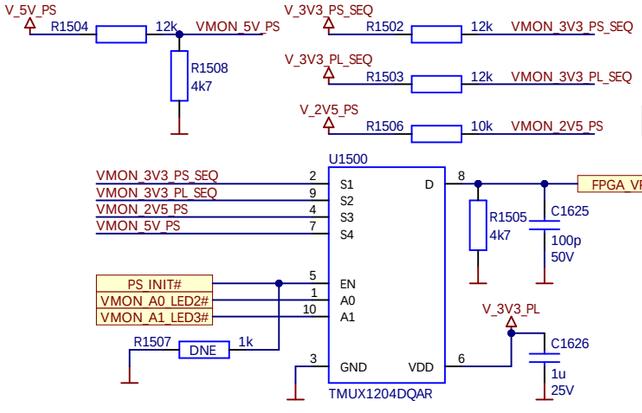
FPGA PL Decoupling

not included in user schematics

FPGA PL IO Decoupling

not included in user schematics

VMON Multiplexer



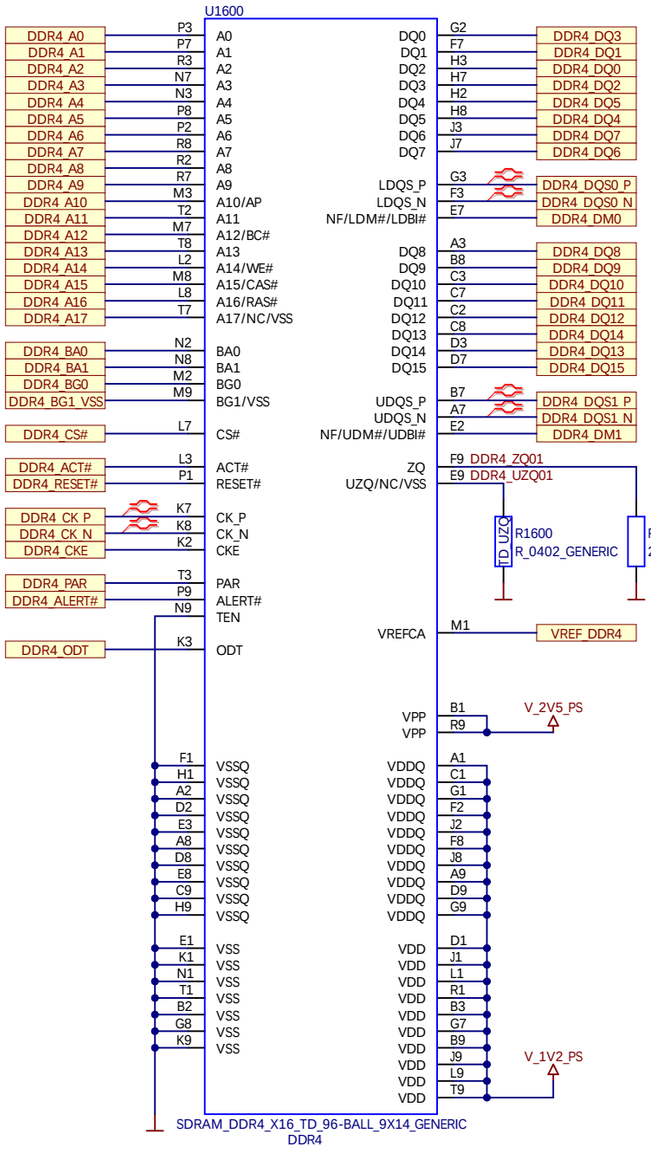
PL System Monitor Power

not included in user schematics

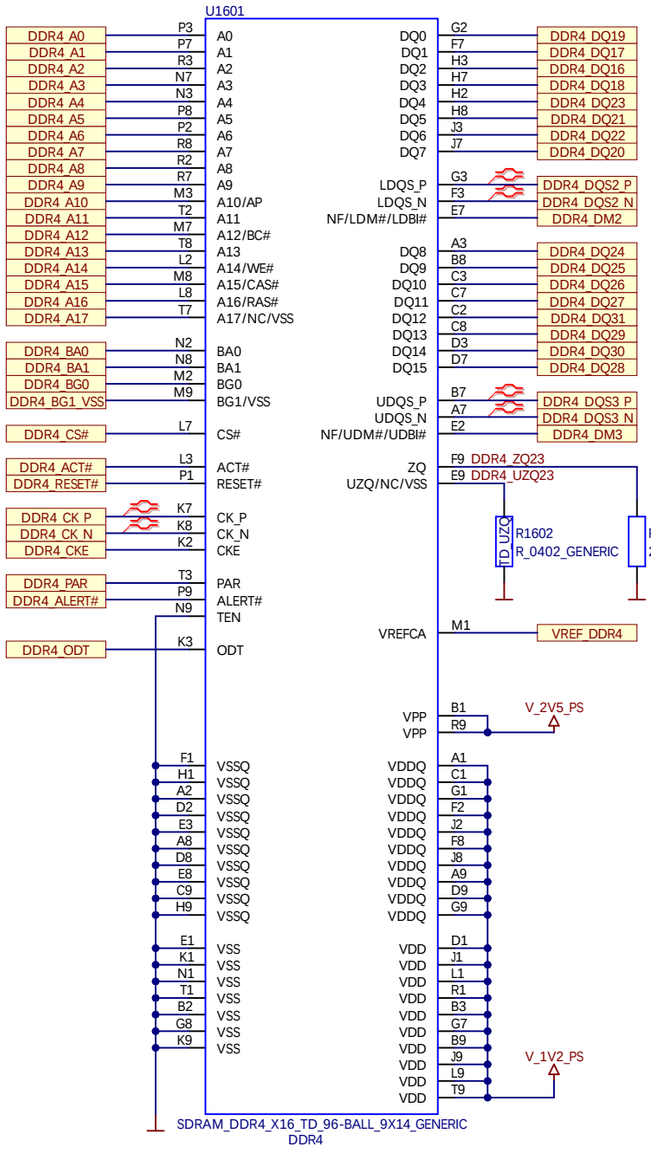
ENCLUSTRA FPGA SOLUTIONS	Project AM-XZU90.PrjPcb	Sheet Name AM-XZU90_P15_FPGA_Supply.SchDoc	Customer No 0000	Revision R2.1	DNE = Do Not Equip	Confidential
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DDR4 SDRAM

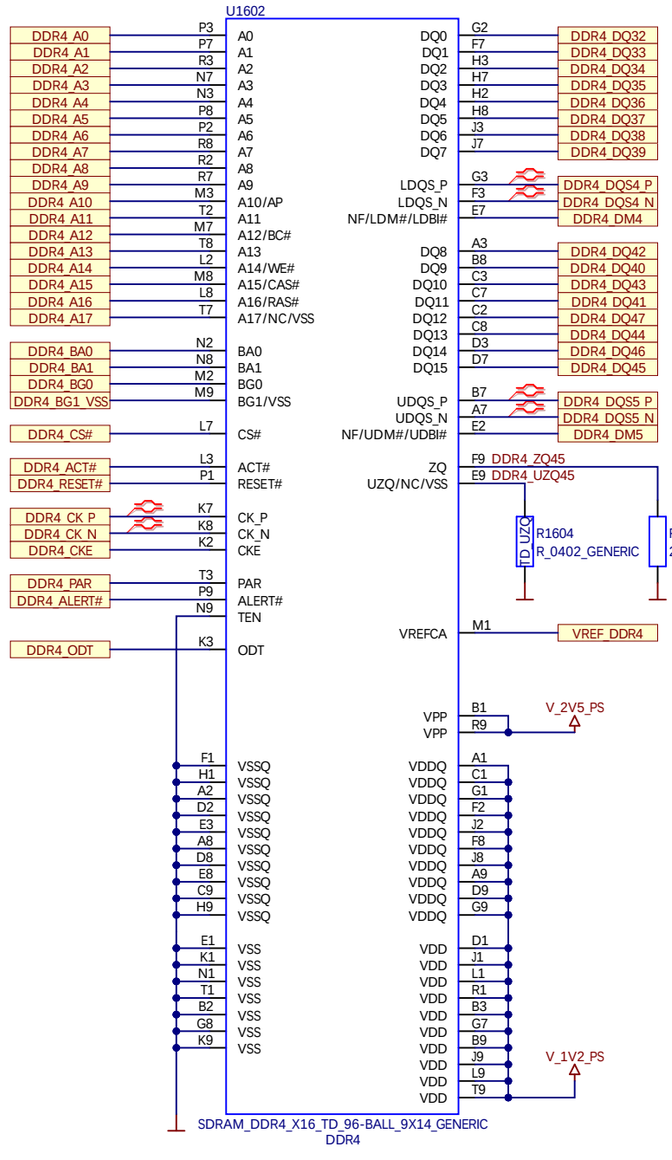
Bytes 0-1



Bytes 2-3



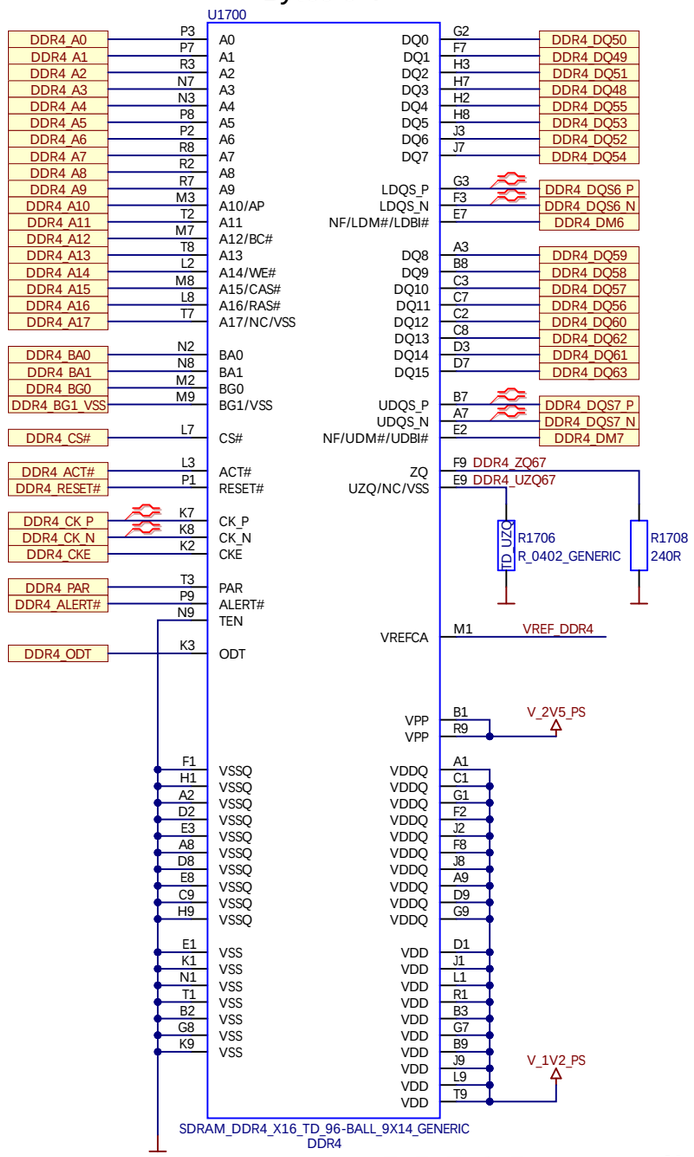
Bytes 4-5



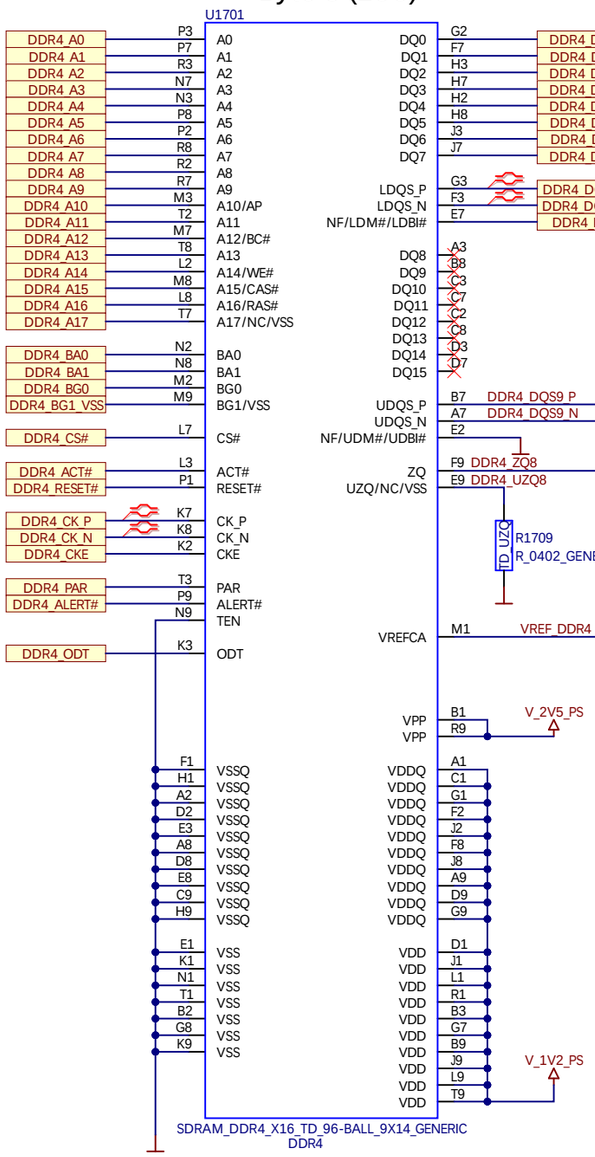
ENCLUSTRA FPGA SOLUTIONS	Project AM-XZU90.PrjPcb	Sheet Name AM-XZU90_P16_SDRAM.SchDoc	Customer No 0000	Revision R2.1	DNE = Do Not Equip	Confidential
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DDR4 SDRAM

Bytes 6-7



Byte 8 (ECC)



DDR4 Termination

not included in user schematics

DDR4 Decoupling

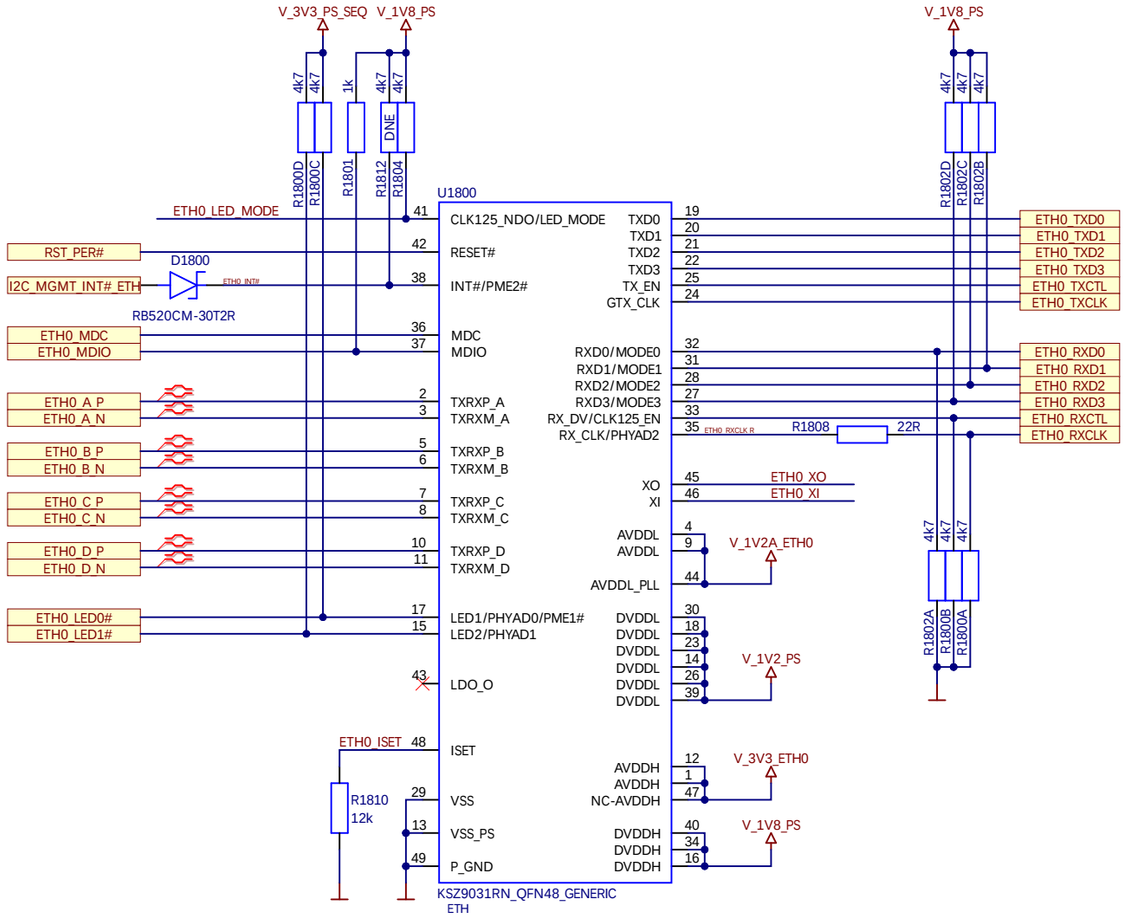
not included in user schematics

DDR4 Decoupling

not included in user schematics

ENCLUSTRA FPGA SOLUTIONS	Project AM-XZU90.PrjPcb	Sheet Name AM-XZU90_P17_SDRAM.SchDoc	Customer No 0000	Revision R2.1	DNE = Do Not Equip	Confidential
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PS Gigabit Ethernet PHY 0

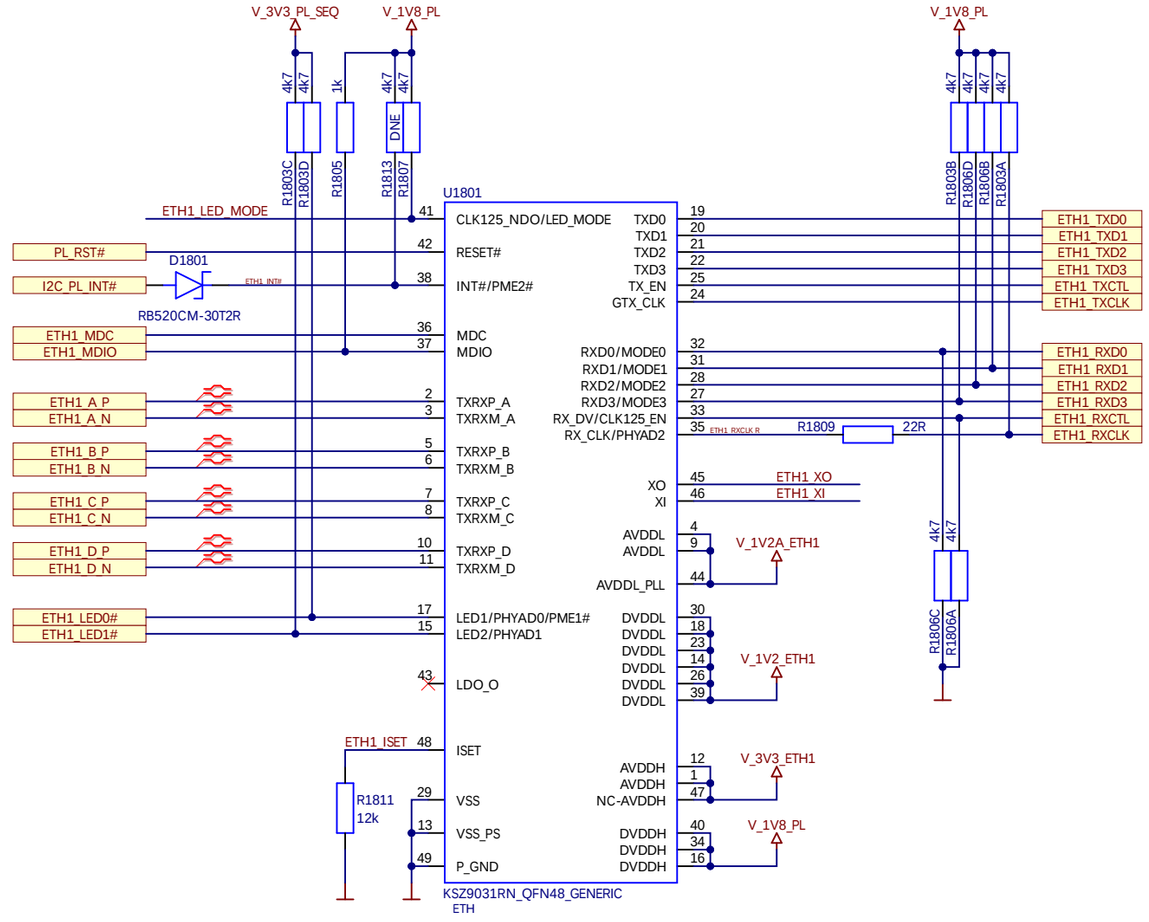


MODE<3:0> = 1110 -> RGMII, all capabilities
LED_Mode Pull up = Single LED

PhyAddress <4:0> =00011 -> 3h

LED1 and LED2 are active low

PL Gigabit Ethernet PHY 1



MODE<3:0> = 1110 -> RGMII, all capabilities
LED_Mode Pull up = Single LED

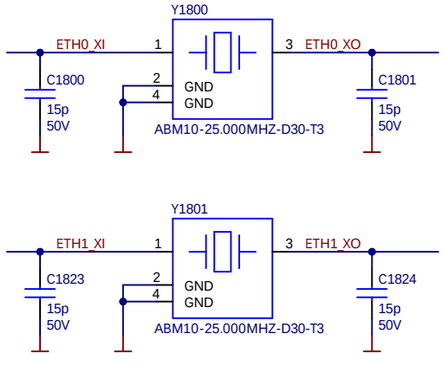
PhyAddress <4:0> = 00111 -> 7h

LED1 and LED2 are active low

PS Ethernet Power Filters & Decoupling

not included in user schematics

Ethernet 25 MHz Crystals

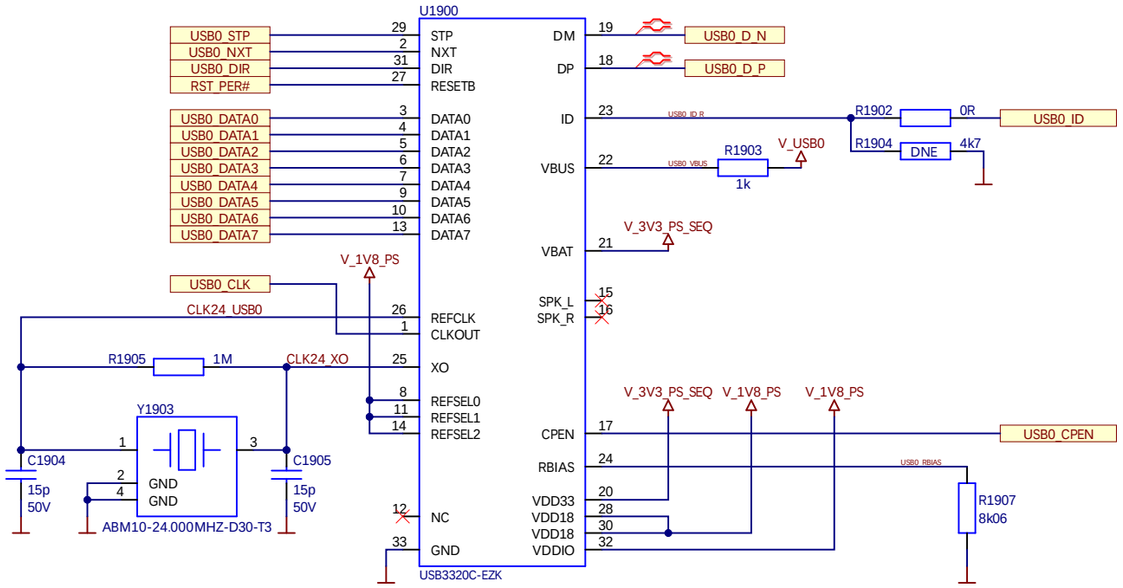


PL Ethernet Power Filters & Decoupling

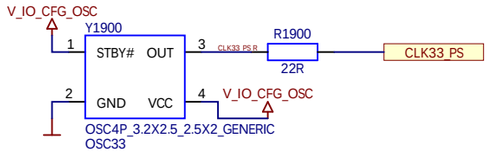
not included in user schematics

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USB 2.0 Device/Host PHY 0



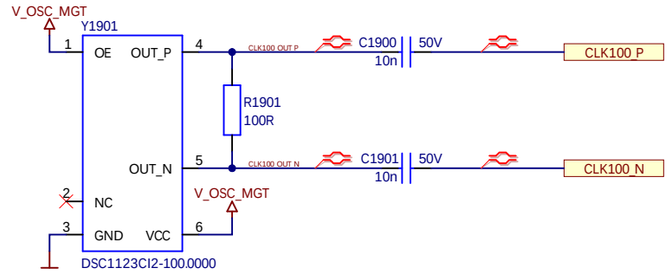
PS Oscillator 33.3 MHz



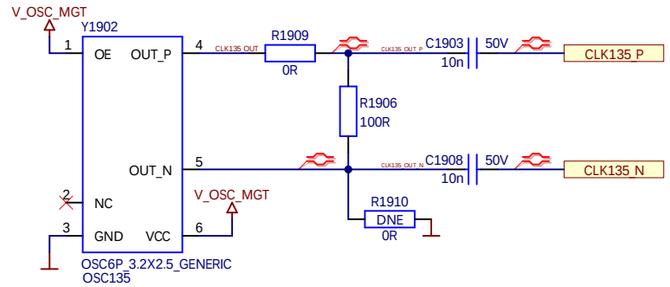
Oscillator Power Filters

not included in user schematics

PS MGT 100 MHz Oscillator



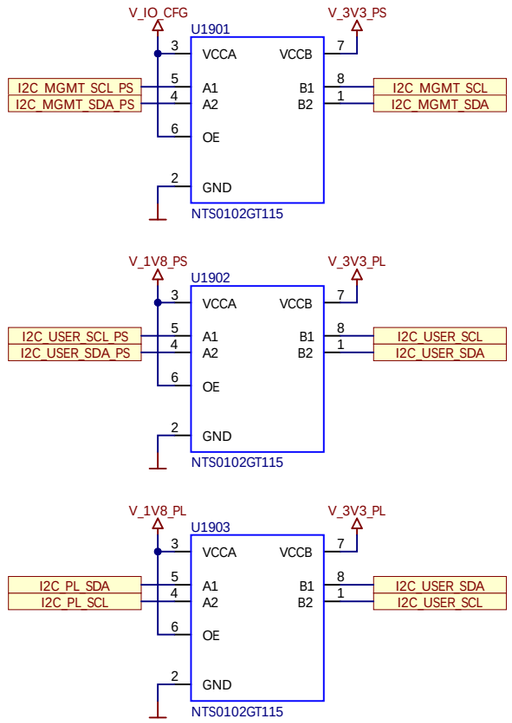
PS MGT 135 MHz Oscillator

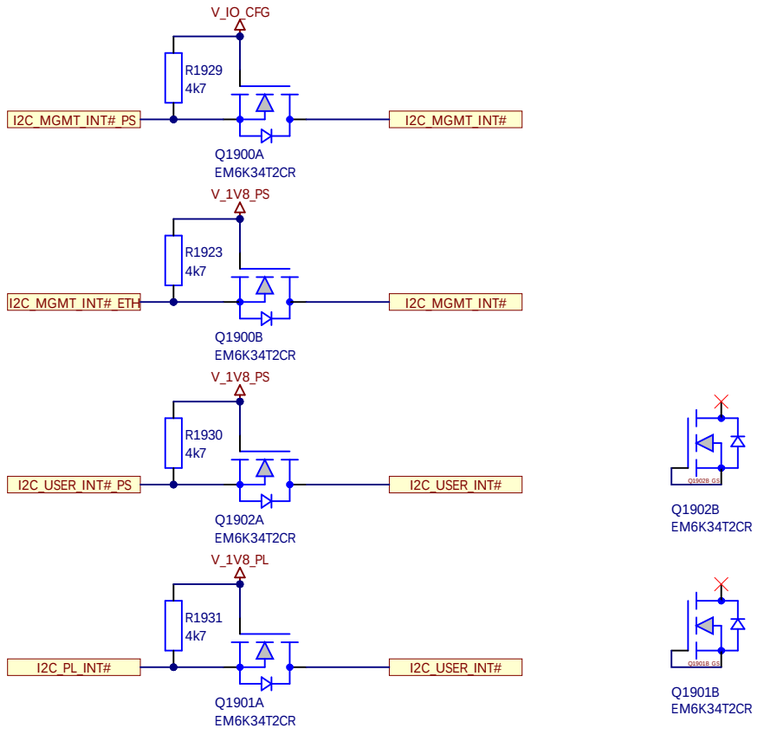


USB Power Decoupling

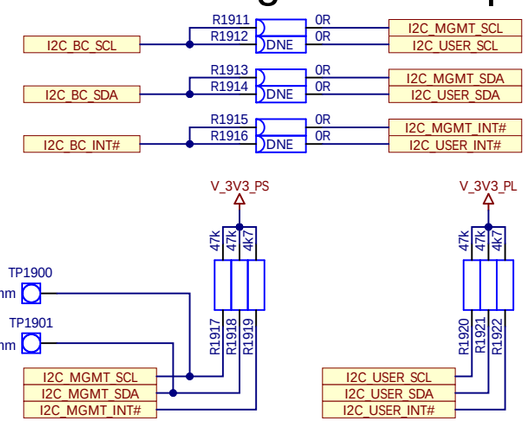
not included in user schematics

I2C Level Shifters





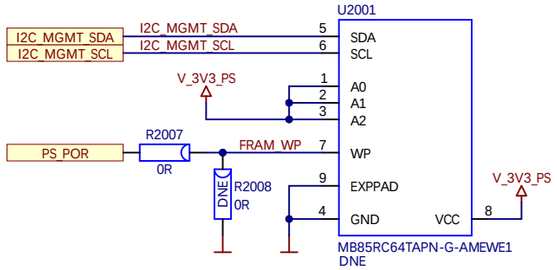
I2C Routing And Pullups



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User FRAM

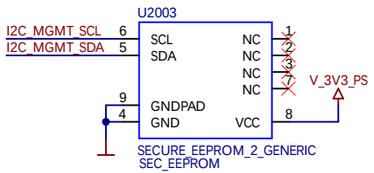
optional



MB85RC64TA I2C Address: 101 0111 Rw (57h)

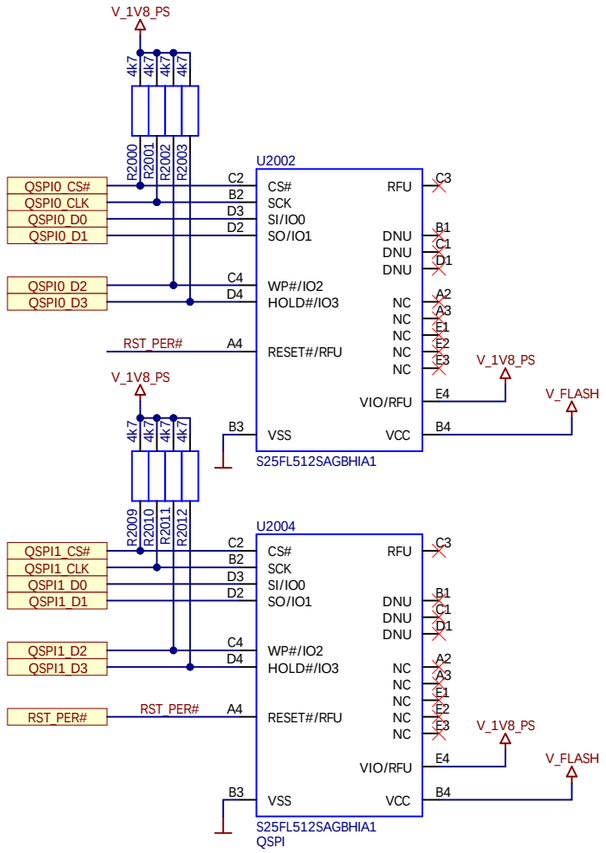
Secure EEPROM

for Enclustra use only



ATSHA204A-MAHDA-T I2C Address: 110 0100 Rw (64h)
ATECC608A-MAHDA-S I2C Address: 110 0100 Rw (64h)

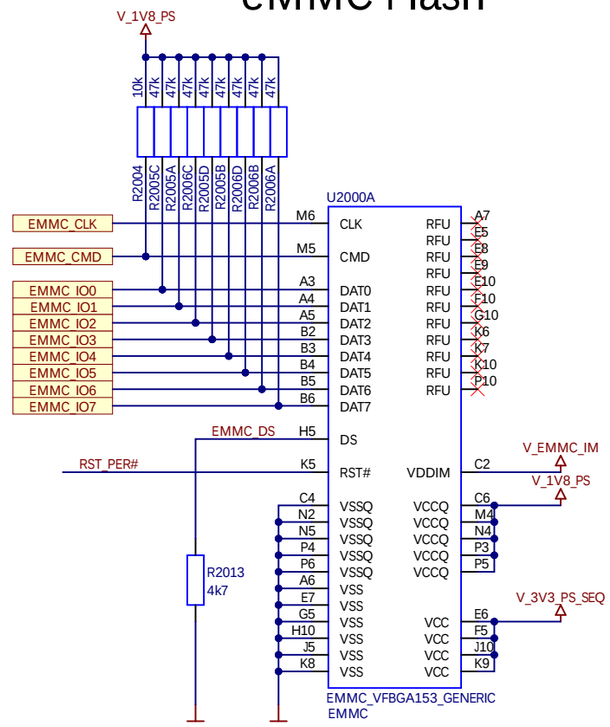
Dual Quad SPI Flash

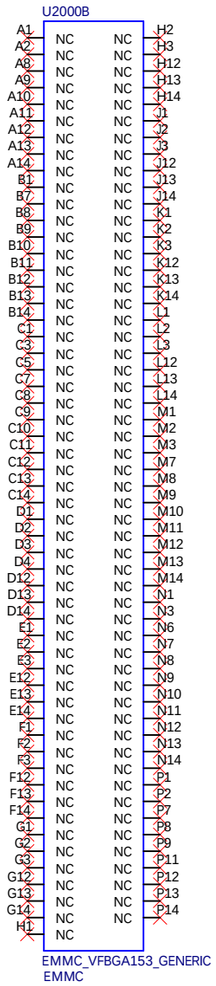


Dual Quad SPI Power

not included in user schematics

eMMC Flash

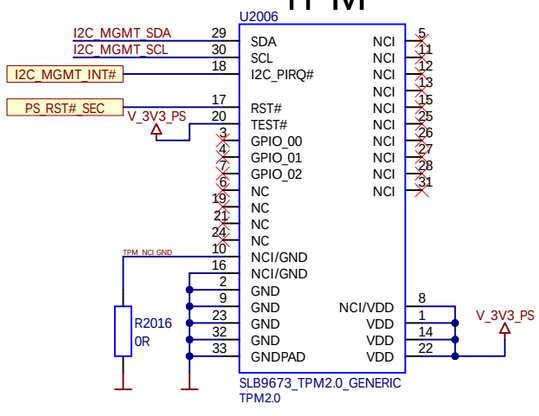




eMMC Power

not included in user schematics

TPM



TPM2.0 I2C Address: 010 1110 Rw (2Eh)

TPM Power

not included in user schematics

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Supported Voltage Ranges

Power Rail	Direction	Range	Tolerance	Description
V_MOD_PL_W V_MOD_PL_X	In	12V	±5%	Module Power for Programmable Logic
V_MOD_PS	In	12V	±5%	Module Power for Processing System
V_BAT	In	2.0 - 5.5V*	-	RTC / BBRAM
V_IO_CFG	In	1.8 - 3.3V	±5%	Configuration Bank
V_IO_B65 V_IO_B66 V_IO_B69 V_IO_B70 V_IO_B71	In	1.0 - 1.8V	±5%	HP IO Banks
V_IO_B91	In	1.8 - 3.3V**	±5%	HD IO Bank
V_3V3_PS_OUT	Out	3.3V	±5%	Power for Carrier
V_3V3_PL	Out	3.3V	±5%	Power for Carrier
V_1V8_PL	Out	1.8V	±5%	Power for Carrier

*Voltage range valid for LDO assembly option (default).
*Voltage range valid for resistor assembly option is 1.2-1.5V.
**Tolerance range valid for 3.3V input option is +3/-5%.

Power Input Filters

not included in user schematics

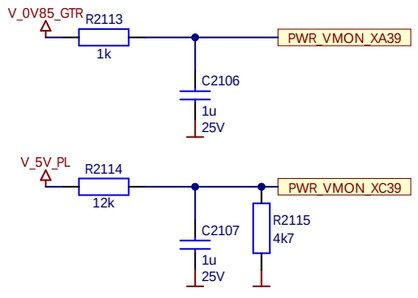
PS/PL Power Sequencing

not included in user schematics

Power Converter Synchronization

not included in user schematics

Voltage Monitoring



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PS Buck Converter 0

I2C-Address = 13h, PMBUS-Address = 43h
not included in user schematics

PS Power Sequencing

not included in user schematics

PS LDO 2.5V 300mA

V_fb = 1.25V
not included in user schematics

PS LDO 0.85V 0.5A

Used for -3E FPGA devices, V_fb = 0.8V
not included in user schematics

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PL Buck Converter 1

I2C-Address = 10h, PMBUS-Address = 40h
not included in user schematics

PL 5V 300mA

not included in user schematics
FS=2.2MHz Vfb=0.8V

PL Power Sequencing

not included in user schematics

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PL Buck Converter 2

I2C-Address = 1Dh, PMBUS-Address = 4Dh
not included in user schematics

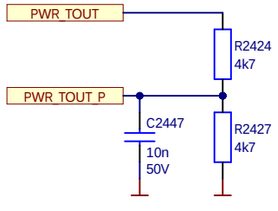
PL Buck Converter 2 Power Stage

not included in user schematics

PL Overtemp/Overcurrent/Fault Reset

not included in user schematics

Power Stage Monitoring



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PL Buck Converter 3

I2C-Address = 1Fh, PMBUS-Address = 4Fh
not included in user schematics

PL Buck Converter 3 Power Stage

not included in user schematics

PS 5V 140mA

not included in user schematics
FS=2.2MHz Vfb=0.8V

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Board Mounting Holes

M2600

AM-MOUNTING-HOLES-MODULE-LARGE

Logos

Z2603

Z2600
94V-0

Z2601

Z2602

Z2604
Andromeda_XZU90

Z2607
E888888

Z2608
0000

Z2605
WW / YY

PCB

P2600

AM-XZU90-B

Texts

Z2609	Z2610	Z2611	Z2612	Z2613	Z2614
S	E	0	1	2	3

Z2615	Z2616	Z2617	Z2618
0	1	2	3

Labels

Z2606

QR 5x5

Z2619
EN000000
SN999999
EN-SN 6X15

PCB Fiducials

FID2600 FID2601 FID2602

FID2603 FID2604 FID2605

Layer Display

Z2620

LAYER DISPLAY 24L

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Component Options

FPGA	SDRAM	EMMC	SEC_EEPROM	L100N	L3U3 (2A)	L1U5 (4A)	OSC33	TD_UZQ	D4_BG1	L220N	Ethernet PHY
Part U800A XCZU17EG-1FFVD1760E EN102422	Part U1600A MT40A1G16TB-062E IT:F EN105414	Part U2000A H26M41204HPR EN101733	Part U2003A ATSHA204A-MAHDA-T EN100827	Part L2300A 100n EN102190	Part L2200A 3u3 EN103135	Part L2202A 1u5 EN103136	Part Y1100A ASDMB-33.333MHZ-LC-T EN100285	Part R1600A 0R EN101361	R1714,R1715 or R1716	Part L2500A 220n EN103125	Part U1800A KSZ9031RNXIA EN101934
Part U800B XCZU17EG-2FFVD1760E EN102978	Part U1600B AS4C512M16D4-75BIN EN104035	Part U2000B H26M52208FPRI EN101865	Part U2003B ATECC608A-MAHDA-S EN103008	Part L2400A 100n EN102190	Part L2200B 3u3 EN103134	Part L2202B 1u5 EN103133	Part Y1100B ASEMB-33.333MHZ-LC-T EN100914	Part R1600B 240R EN101382		Part L2500B 220n EN103129	Part U1800B KSZ9031RNXIC EN104260
Part U800C XCZU19EG-2FFVD1760I EN102646	Part U1601A MT40A1G16TB-062E IT:F EN105414	Part U2000C EMMC16G-IB29-PZ90 EN102591		Part L2300B 120n EN103128	Part L2200C 3u3 EN103132	Part L2202C 1u5 EN103131	Part Y1900A ASDMB-33.333MHZ-LC-T EN100285	Part R1602A 0R EN101361		Part L2500C 220n EN103127	Part U1801A KSZ9031RNXIA EN101934
	Part U1601B AS4C512M16D4-75BIN EN104035	Part U2000D FEMDRW016G-88A43 EN104303		Part L2400B 120n EN103128	Part L2201A 3u3 EN103135	Part L2203A 1u5 EN103136	Part Y1900B ASEMB-33.333MHZ-LC-T EN100914	Part R1602B 240R EN101382			Part U1801B KSZ9031RNXIC EN104260
	Part U1602A MT40A1G16TB-062E IT:F EN105414	EN101733 can only be used in product model with 0 to 85°C operating temperature range		Part L2300C 100n EN103124	Part L2201B 3u3 EN103134	Part L2203B 1u5 EN103133		Part R1604A 0R EN101361			
	Part U1602B AS4C512M16D4-75BIN EN104035			Part L2400C 100n EN103124	Part L2201C 3u3 EN103132	Part L2203C 1u5 EN103131		Part R1604B 240R EN101382			
	Part U1700A MT40A1G16TB-062E IT:F EN105414			Part L2300D 100n EN105316	Part L2401A 3u3 EN103135	Part L2402A 1u5 EN103136		Part R1706A 0R EN101361			
	Part U1700B AS4C512M16D4-75BIN EN104035			Part L2400D 100n EN105316	Part L2401B 3u3 EN103134	Part L2402B 1u5 EN103133		Part R1706B 240R EN101382			
	Part U1701A MT40A1G16TB-062E IT:F EN105414				Part L2401C 3u3 EN103132	Part L2402C 1u5 EN103131		Part R1709A 0R EN101361			
	Part U1701B AS4C512M16D4-75BIN EN104035				Part L2501A 3u3 EN103135	Part L2403A 1u5 EN103136		Part R1709B 240R EN101382			
					Part L2501B 3u3 EN103134	Part L2403B 1u5 EN103133					
					Part L2501C 3u3 EN103132	Part L2403C 1u5 EN103131					
						Part L2502A 1u5 EN103136					
						Part L2502B 1u5 EN103133					
						Part L2502C 1u5 EN103131					
						Part L2503A 1u5 EN103136					
						Part L2503B 1u5 EN103133					
						Part L2503C 1u5 EN103131					

Component Assembly

Enclustra Product	FPGA	SDRAM	EMMC	SEC_EEPROM	L100N	L2U2 (2.6A)	L1U (4.6A)	OSC33	TD_UZQ	D4_BG1	L220N	Ethernet PHY
Variant EN108102:0 AM-XZU90-17EG-1E-D12E-R2.1	Option EN108102:1 FPGA EN102422	Option EN108102:2 DDR4 EN104035	Option EN108102:3 EMMC EN104303	Option EN108102:4 SEC_EEPROM EN100827	Option EN108102:5 L100N EN105316	Option EN108102:6 L3U3 EN103135	Option EN108102:7 L1U5 EN103136	Option EN108102:8 OSC33 EN100914	Option EN108102:9 TD_UZQ EN101361	Option EN108102:10 D4_BG1 DNE	Option EN108102:11 L220N EN103125	Option EN108102:12 ETH EN101934
Variant EN108103:0 AM-XZU90-17EG-2E-D12E-R2.1	Option EN108103:1 FPGA EN102978	Option EN108103:2 DDR4 EN104035	Option EN108103:3 EMMC EN104303	Option EN108103:4 SEC_EEPROM EN100827	Option EN108103:5 L100N EN105316	Option EN108103:6 L3U3 EN103135	Option EN108103:7 L1U5 EN103136	Option EN108103:8 OSC33 EN100914	Option EN108103:9 TD_UZQ EN101361	Option EN108103:10 D4_BG1 DNE	Option EN108103:11 L220N EN103125	Option EN108103:12 ETH EN101934
Variant EN108104:0 AM-XZU90-19EG-2I-D12E-R2.1	Option EN108104:1 FPGA EN102646	Option EN108104:2 DDR4 EN104035	Option EN108104:3 EMMC EN104303	Option EN108104:4 SEC_EEPROM EN100827	Option EN108104:5 L100N EN105316	Option EN108104:6 L3U3 EN103135	Option EN108104:7 L1U5 EN103136	Option EN108104:8 OSC33 EN100914	Option EN108104:9 TD_UZQ EN101361	Option EN108104:10 D4_BG1 DNE	Option EN108104:11 L220N EN103125	Option EN108104:12 ETH EN101934
Variant EN108105:0 AM-XZU90-19EG-2I-D13E-R2.1	Option EN108105:1 FPGA EN102646	Option EN108105:2 DDR4 EN105414	Option EN108105:3 EMMC EN104303	Option EN108105:4 SEC_EEPROM EN100827	Option EN108105:5 L100N EN105316	Option EN108105:6 L3U3 EN103135	Option EN108105:7 L1U5 EN103136	Option EN108105:8 OSC33 EN100914	Option EN108105:9 TD_UZQ EN101361	Option EN108105:10 D4_BG1 DNE	Option EN108105:11 L220N EN103125	Option EN108105:12 ETH EN101934

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Component Options

OSC135	TPM2.0	QSPI									
Part Y1902A DSC1123CI2-135.0000 EN105426 Part Y1902B SIT9121AI-2B1-33E135.000 EN104152 Part Y1902C SJK-S3D5000M 3C04 20B 8 EN104153 For single-ended oscillators: R1909 = 240R, R1910 = 0R For differential oscillators: R1909 = 0R, R1910 = DNE	Part U2006A SLB9673XU20FW2613XTMA1 EN106626	U2004									

Component Assembly

OSC135	TPM2.0	QSPI									
Option EN108102:13 OSC135 EN105426 Option EN108103:13 OSC135 EN105426 Option EN108104:13 OSC135 EN105426 Option EN108105:13 OSC135 EN105426	Option EN108102:14 TPM2.0 EN106626 Option EN108103:14 TPM2.0 EN106626 Option EN108104:14 TPM2.0 EN106626 Option EN108105:14 TPM2.0 EN106626	Option EN108102:15 QSPI Option EN108103:15 QSPI Option EN108104:15 QSPI Option EN108105:15 QSPI									

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